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Vanadium-Zeolite Interactions in Fluidized Cracking Catalysts

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Vanadium–Zeolite Interactions in Fluidized Cracking Catalysts*

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I. INTRODUCTION

Metal levels on equilibrium fluidized cracking catalysts (FCC) have been increasing through the years (Fig. 1), indicating that the study of the physicochemical properties of metal resistant FCC has remained an important topic to refiners [1]. After the crude oil price collapse of 1986, Ni levels on equilibrium FCC have remained fairly constant whereas V levels have somewhat decreased owing to the greater availability (and lower cost) of high-quality crudes; Fig. 2. At the metals levels shown in Fig. 1, Ni and V catalyze the undesirable cracking of hydrocarbons in the gasoline boiling range, thus increasing coke and hydrogen yields. Vanadium-induced activity losses can be easily circumvented with the proper catalyst management and increased makeup rates. However, should refiners in the future undertake the cracking of residual oils or crack certain readily available crudes from Venezuela and Mexico, metals (especially V) will probably increase well above the levels shown in Fig. 1 and novel FCC will have to be found to negate the severe effects of V on cracking activity.

It is well known that Ni (like Fe) does not quench cracking activity even when present at the 2–3% level [2–9]. In contrast, above 0.5% V loading, catalyst properties deteriorate rapidly and at 1% V a near total loss of surface area and (useful) cracking activity is expected [7, 10]. At microactivity test conditions, it has been shown that addition of diluents capable of selectively sorbing metal contaminants from gas oils can form dual function cracking catalysts (DFCC) mixtures that retain most of their useful cracking activity even in the presence of 1.0–1.5% V [10–13]. The deleterious effects of V on cracking activity can be inhibited also by incorporating metal traps, such as sepiolite [14], titanates [15, 16], and sacrificial zeolites [17, 18] into the FCC matrix.

There is another reason that justifies interest in studying FCC contaminated with metal levels above values currently found on equilibrium FCC. The

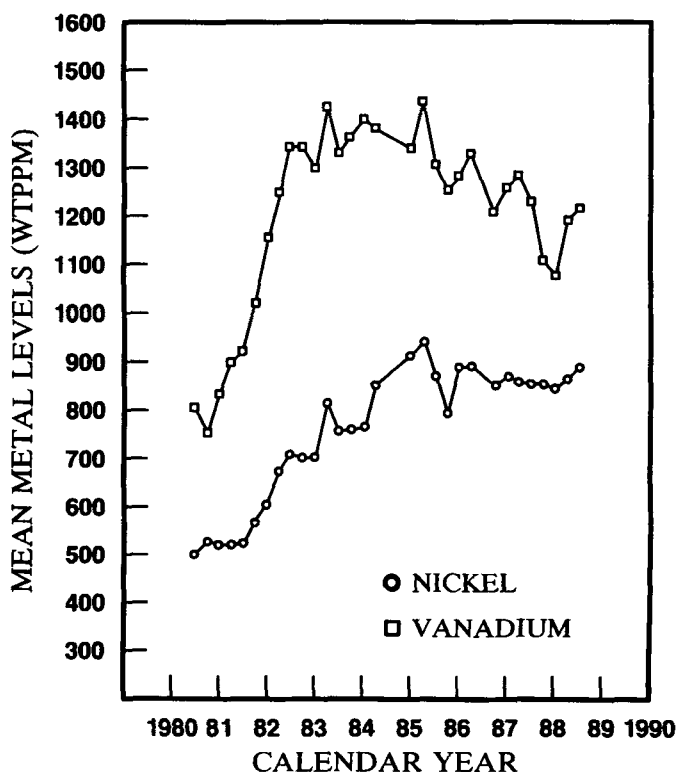


FIG. 1. Metal level trends in equilibrium fluid cracking catalysts (FCC). Data based on Davison analysis of samples from 96% of fluid units in the USA and Canada.

metal loadings shown in Fig. 2 would preclude (or greatly complicate) the study of metals-surface interaction in FCC by some of the more readily available characterization techniques such as x-ray diffraction (XRD), x-ray photoelectron spectroscopy (XPS), nuclear magnetic resonance (NMR), luminescence, laser raman spectroscopy (LRS), and secondary ion mass spectrometry (SIMS). Thus, most of the research has been conducted with materials containing 0.5% to 5.0% metals deposited using rapid methods of artificially contaminating catalysts that employ solutions of Ni or V naphthenates in benzene (or toluene) [19, 20]. The catalysts are calcined (500–600°C)

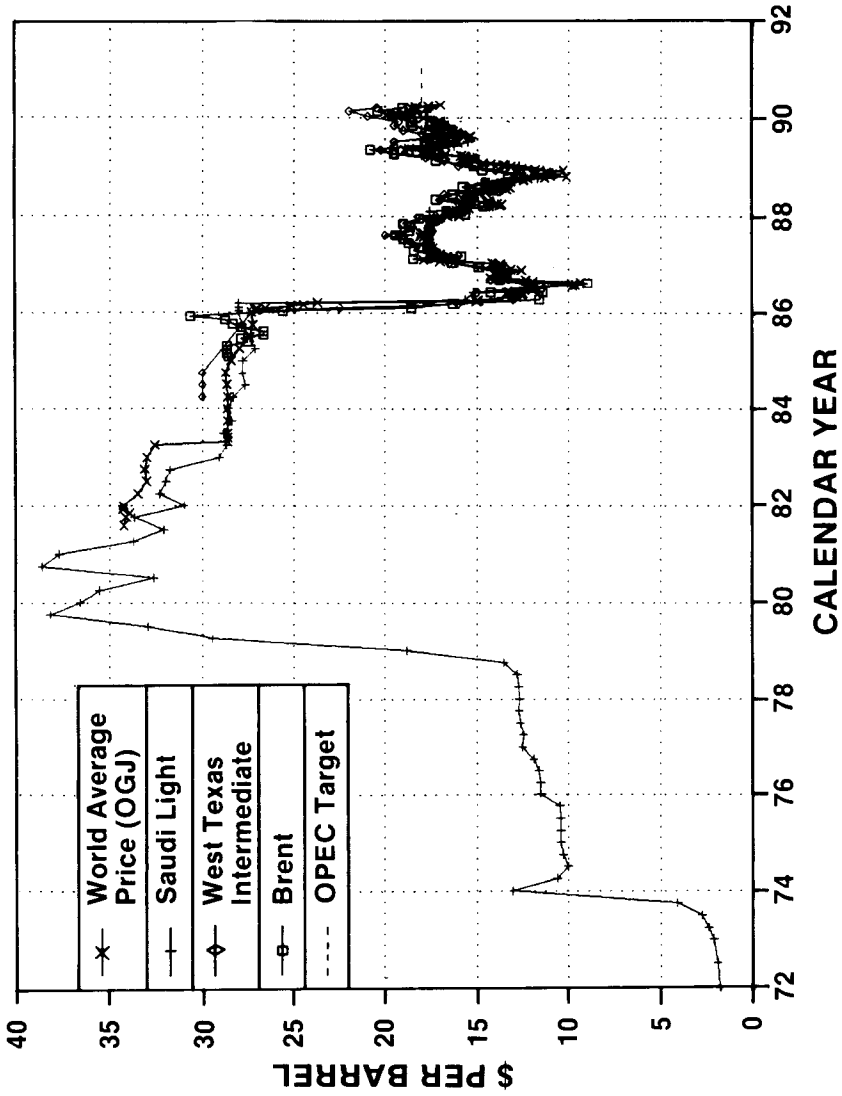


FIG. 2. Crude oil prices for the 1970–1990 period.

in air to remove the organics and then (to reduce their properties to those found in equilibrium FCC) exposed to a flow of steam at 700–800°C for a 5- to 10-h period.

From materials thus prepared, luminescence and electron paramagnetic resonance (EPR) data [21, 22] have indicated that after V loading a FCC cracking component (a zeolite with the faujasite structure) or an FCC matrix (an aluminosilicate gel of the type used in FCC preparations) VO^{2+} ions are formed. Following the oxidative decomposition of the naphthenate (540°C/10 h in air), V on the gel is oxidized mainly to V_2O_5 whereas in the zeolite, it remains present mainly as vanadyl cations. Only after steaming is V present on the zeolite, mostly as a V^{5+} specie [21–23].

Steam-aging (at 760°C/5 h) causes the collapse of a V-loaded zeolite Y by two different mechanisms. In HY crystals [free of rare earth (RE) cations], after lattice collapse, the formation of V_2O_5 , mullite, and silica (tridymite) is observed [11]. In the presence of RE ions, the faujasite structure collapse with formation of rare earth vanadates (REVO_4) [11]; The formation of these compounds has been proposed (or observed) by several other workers [25–29]. It is the purpose of this paper to review and to further examine the interaction of V with zeolites (HY or CREY) usually employed as cracking components in FCC formulations.

II. EXPERIMENTAL

A. Catalyst Preparation and Testing

The sample of calcined rare earth (RE) exchanged zeolite Y (CREY) and the high-activity cracking catalyst (GRZ-1) used in the present study were obtained from the Davison Company. Davison's GRZ-1 is an FCC that contains an estimated 35% CREY and after steam-aging has a BET surface area of 161 m^2/g . The CREY sample had a $\text{SiO}_2/\text{Al}_2\text{O}_3$ ratio of 5.0; contained 7.6% Ce_2O_3 , 4.0% La_2O_3 , 2.8% Nd_2O_3 , and 0.9% Pr_2O_3 ; and had a BET surface area of 749 m^2/g . A residual 3.5% Na_2O is present in these crystals.

The low (0.16%) Na_2O REHY sample had BET surface area of 560 m^2/g and $\text{SiO}_2/\text{Al}_2\text{O}_3 = 4.7$. The crystals contained 0.91% Ce_2O_3 , 5.2% La_2O_3 , 2.0% Nd_2O_3 and 0.71% Pr_2O_3 . The HY sample (Linde LZY-82) had a $\text{SiO}_2/\text{Al}_2\text{O}_3$ ratio of 5.4, its BET surface area was 761 m^2/g , and it contained 0.14% Na_2O . LaY (containing 8.1% La_2O_3) was obtained by reacting this HY with 1 M LaCl_3 solutions (prepared using $\text{LaCl}_3 \cdot 7\text{H}_2\text{O}$ crystals, from GFS Chemicals). The silicon-enriched HY crystals used (Linde LZ-210) had

BET surface area of 746 m²/g and a SiO₂/Al₂O₃ ratio of 6.5, and contained 0.18% Na₂O.

Catalyst evaluation was performed with a microactivity test (MAT) using conditions described elsewhere [9]. Conversions are on a vol% fresh feed (FF) basis and have been defined as $[V_f - V_p/V_f] \times 100$, where V_f is the volume of feed and V_p is the volume of product with bp > 204°C.

Solutions of vanadyl naphthenate in benzene were used to metal load the various materials according to an established procedure [19]; the naphthenate was obtained from Pfaltz and Bauer, Inc., and contained 2.9 wt% V. The metal-loaded materials were first air-dried for 10 h, slowly brought to 540°C (in flowing air), and then kept at this temperature for 10 h. Steam-aging was accomplished by passing steam at 760°C over the calcined catalysts for 5 h.

B. Vanadium Analysis

Vanadium in several aqueous extracts was determined by flame atomic emission spectrometry. The extracts were diluted with distilled water and potassium chloride was added as an ionization buffer. Atomic emission at 437.92 nm was measured after aspirating the solutions into a fuel-rich nitrous oxide-acetylene flame. Quantitation was accomplished by comparing the sample emission intensities with a calibration curve which was prepared from aqueous standard vanadium solutions. A polynomial regression was used to fit the vanadium standards to the calibration curve.

C. Powder Diffraction Measurements

Powder diffraction measurements were obtained with a Siemens D-500 diffractometer at a scan rate of 0.01°/step using 3 sec/step monochromatic CuK_α radiation; CaF₂ was used as an internal standard for angle calibration. Peak positions of the mullite samples were determined using computer programs supplied by the instrument vendor and corrected to JCPDS CaF₂ positions. Lattice parameters and cell volume were determined using a PC version of the Appleman refinement program as supplied by Benoit. Only 10 to 15 reflections were used in the refinement due to the somewhat poor crystallinity and weak reflections present in some samples.

D. Raman Spectroscopy

Raman spectra were recorded on a Spex Ramalog 1403 spectrometer (Spec Industries, Metuchen, NJ) equipped with a cooled RCA GaAs photomultiplier

tube (CA 31034-02). The 4880 Å line of a model 165 Ar⁺ laser (Spectra Physics, Mountain View, CA) was used to generate Raman scattered light. The laser power impinging on the sample was limited to approximately 50 mW. All spectra were recorded with a spectral resolution of 5 cm⁻¹ (350 μm slit width). Signal pulses from the photomultiplier tube were passed through a model 1182 amplifier/discriminator (Princeton Applied Research, Princeton, NJ) and counted by a Nicolet 1280 data system (Nicolet Instrument Corp., Madison, WI). Typically, 50–100 scans per sample were averaged in order to obtain spectra with adequate signal-to-noise ratios. A data point was acquired every 0.7 cm⁻¹. Samples were prepared for Raman measurements by grinding the catalysts in an agate mortar and pestle. Samples containing KNO₃ (the internal standard) were prepared by grinding about 90% catalyst and 10% KNO₃ mixtures. The powdered samples were then pressed into 1 cm diameter wafers under a pressure of approximately 1000 psi. Each wafer was mounted in a spinning sample holder and placed in the sample chamber of the spectrometer. The angle between the incident laser beam and the sample surface was 30°; samples were spun at approximately 500 rpm to minimize thermal modifications.

E. Si and Al NMR Measurements

Silicon-29 NMR spectra were recorded at 53.7 MHz on an IBM AF-270 FTNMR spectrometer equipped with Doty Scientific MAS solids probe. Samples were spun in cylindrical 7 mm alumina (sapphire) rotors equipped with vespel caps at 3.0 to 3.5 kHz. Since neither high-power proton decoupling nor cross-polarization (CP) was employed, the observed signal intensities are uncorrected for the possible presence of Si—OH groups. A 60° pulse (6 μs) and 5 sec recycle delay was used in all cases. 2K data points were acquired during a 51 msec acquisition time (corresponding to 20 kHz spectral width) and zero filled to 4K data points with apodization equivalent to 20- to 50-Hz line broadening applied prior to Fourier transformation. Chemical shifts were referenced to external TMS (taken as 0.0 ppm) by sample exchange using a secondary standard (set to -108.2 ppm).

Experimental Silicon-29 NMR signals were deconvoluted into Gaussian components using the LINESIM program (courtesy of Dr. Peter Barron, Griffith University, Natham, Australia) that was written for an ASPECT-3000 computer. The best fitting simulated spectra were obtained using an iterative simplex routine. The Si/Al ratios were calculated from the derived line intensities using the relation [31]:

$$\text{Si/Al} = I_{\text{TOT}} / \sum_{n=1}^4 0.25 n I_{\text{Si}[n\text{Al}]}$$

where I_{TOT} is the total intensity of the spectrum and $I_{\text{Si}[n\text{Al}]}$ is the intensity contributions from Si atoms with $n\text{Al}$ neighbors in their second coordination sphere.

The same spectrometer was also used to record aluminum-27 MAS NMR spectra at 70.4 MHz. Typical scan conditions involved 18° (1 μsec) pulse with a recycle delay of 2 secs to obtain near quantitative results. In calculating Al(VI)/Al(IV) ratios, it was assumed that spinning side bands (SSB) were of equal intensities. 2K data points were acquired over 31 msec acquisition time (corresponding to a sweep width of 33.3 kHz). Data was zero filled to 4K and exponential apodization equivalent to 50 Hz line broadening applied before Fourier transformation. Left shifting was not performed on the acquired data. Samples were spun in 7 mm zirconia rotors fitted with vespel end caps, at 3.0 to 3.5 kHz. Chemical shifts reported are referenced to external aqueous $\text{Al}^{3+}(\text{H}_2\text{O})_6$ (set to 0.0 ppm).

F. Vanadium-51 NMR

Spectra at a resonance frequency of 23.51 MHz were obtained at ambient temperature using a JEOL FX-900 FTNMR spectrometer equipped with a 10 mm multinuclear probe and an external 2H reference lock frequency. The data were acquired over a spectral width 20 kHz using 4K data points, a 45 pulse, and a recycle delay of 10 msec. An exponential multiplication equivalent to a line broadening of 20–30 Hz was applied prior to Fourier transformation. Rolling baselines were often observed, requiring the use of baseline correction routine available with the standard JEOL system software. The chemical shifts are reported with respect to an external VOCl_3 standard set to 0.0 ppm.

III. RESULTS AND DISCUSSION

A. NMR Characterization

The silicon-29 MASNMR spectra of the low Na_2O HY crystals (Linde LZY-82) after calcination in air at 540°C for 10h exhibit resonances at -107.8 ppm, -102.1 ppm, -96.0 ppm, and -91.1 ppm, corresponding to $\text{Si}[\text{OAl}]$, $\text{Si}[\text{1Al}]$, $\text{Si}[\text{2Al}]$ and $\text{Si}[\text{3Al}]$ units, respectively, Fig. 3—trace A; the computed Si/Al ratio from peak intensities is 5.3. After steaming, dealumination occurs and

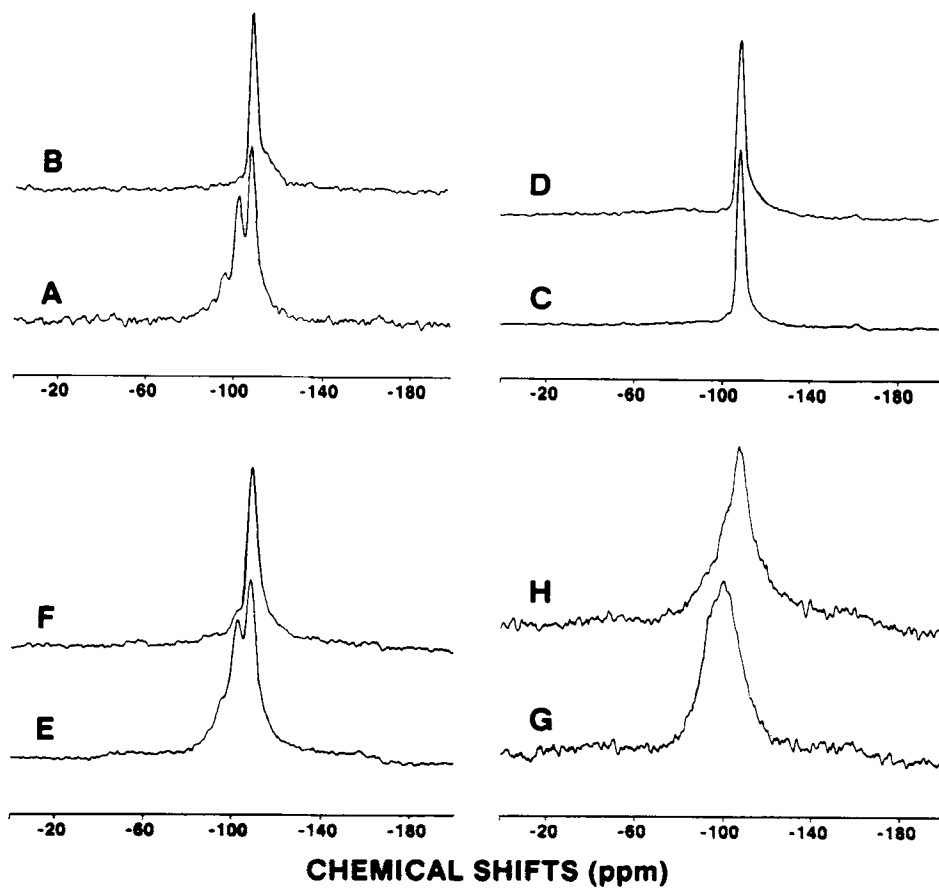


FIG. 3. Silicon-29 NMR Spectra from HY (Linde's LZ-82), silicon-enriched HY (Linde's LZ-210), REHY, and CREY crystals after calcination: traces A, C, E, and G; and after steam-aging: traces B, D, F, and H.

only one resonance at -108.6 ppm is seen in the corresponding spectrum; the shoulder at about -115 ppm suggest formation of some SiO_2 resulting from lattice degradation, Fig. 3—B. By contrast, the spectrum of the silicon-enriched HY crystals (Linde's LZ-210) does not change appreciably during steam aging and only one peak at ~ 109 ppm, owing to the presence of $\text{Si}[\text{OAl}]$ units, can be seen, Fig. 3—C, 3—D. The silicon-29 NMR spectrum of calcined REHY is similar to that obtained for the HY crystals under study, Fig. 3—E and 3—F. Notwithstanding the presence of RE ions, the signal intensities in Fig. 3—E reflect a framework Si/Al ratio of 4.8. Dealumination in the calcined samples is responsible for computed Si/Al ratios almost twice as large as those measured from chemical analysis. As expected, steam-induced dealumination cause the resonances in the -103 ppm to -90 ppm range to disappear, thus generating a spectrum similar to that of the two steam-aged HY crystals, Fig. 3. The spectrum of calcined CREY is featureless owing to the paramagnetic broadening effects of the high ($\sim 15\%$ Re_2O_3) rare earths levels present in these crystals, Fig. 4—G. After steaming the peak maxima at -100.9 ppm shifts to -107.3 ppm, indicating dealumination, Fig. 4—H.

The corresponding aluminum-27 NMR spectra of these zeolites are shown in Fig. 4. As seen in the ^{29}Si NMR spectra, calcination of $540^\circ\text{C}/10$ h in air induce some dealumination in all the samples and the presence of RE ions seems to improve lattice stability. The spectra in Fig. 4 contain an intense asymmetric peak in the 55- to 57-ppm region associated with framework Al atoms and one near 0 ppm associated with extra lattice $\text{Al}(\text{VI})$. The presence of extra framework Al in the calcined samples is best observed by recording ^{27}Al NMR spectra at 400 MHz (instead of 70.4 MHz) and by spinning the samples at 7.0 kHz (instead of 3.0 kHz); see Fig. 4—E and 4—G. In the calcined HY zeolite studied, a shoulder near 30 ppm can now be observed. This shoulder becomes a well-resolved peak in the NMR spectra of the calcined Si-enriched HY crystals, Fig. 4—G. The calcined HY-type crystals seem to contain the most extra framework Al, Fig. 4—A, 4—C, 4—E, and 4—G. Lesser amounts of $\text{Al}(\text{VI})$ are detected in the RE-containing zeolites with the CREY crystals showing the least amounts of $\text{Al}(\text{VI})$, Fig. 4—I—4—M.

Steaming ($760^\circ\text{C}/10$ h) increases dealumination and a contraction of the crystal's unit cell dimensions is observed:

Treatment	LZ-210	LZY-82	REHY	CREY
Calcination	24.303	24.480	24.412	24.695
Steaming	24.260	24.235	24.407	24.599

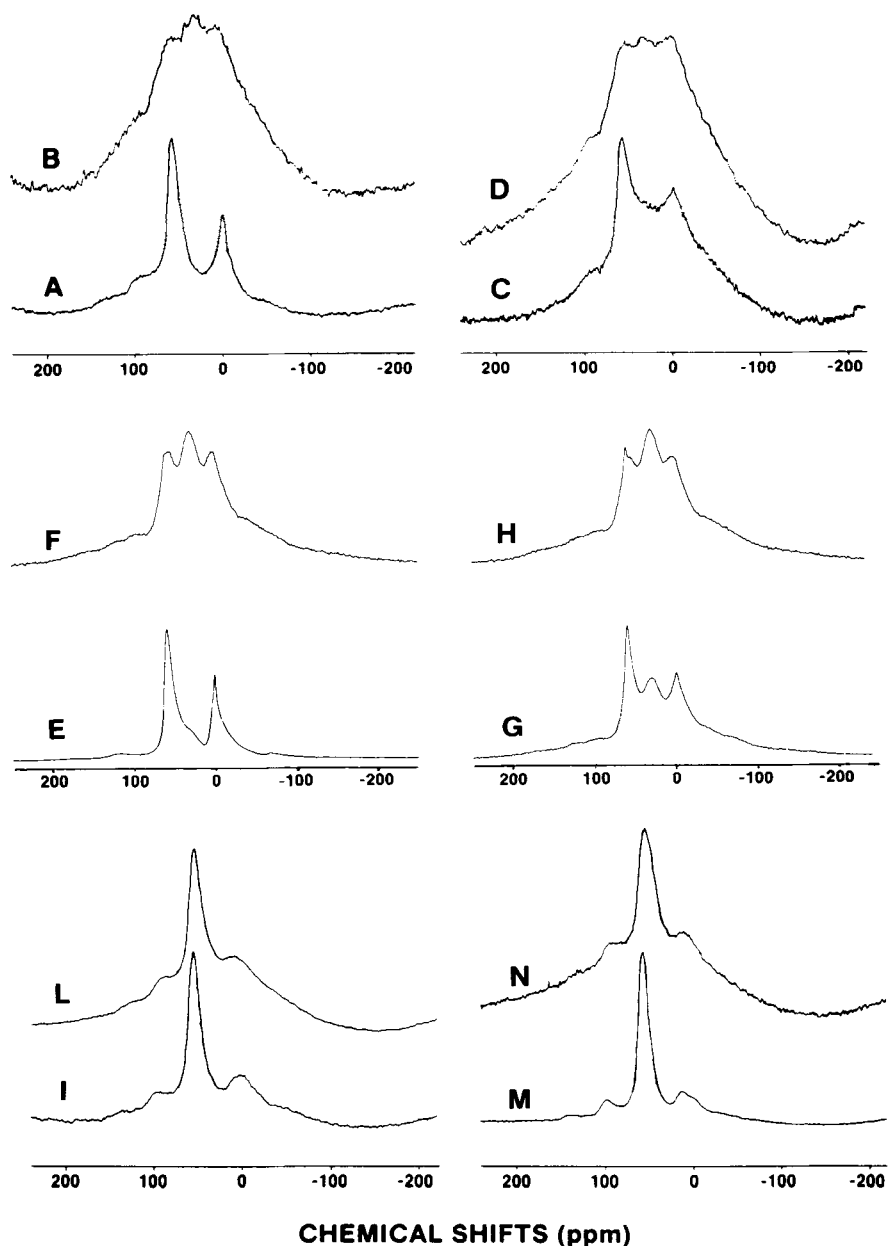


FIG. 4. Aluminum-27 NMR spectra at 70.4 MHz from HY (linde's LZ-82), silicon-enriched HY (Linde's LZ-210), REHY, and CREY crystals after calcination: A, C, I, and M; and after steam-aging: B, D, L, and N. Spectra at 400 MHz for Linde's LZ-82 and LZ-210 after calcination and after steam-aging are shown in E, G, F, and H, respectively.

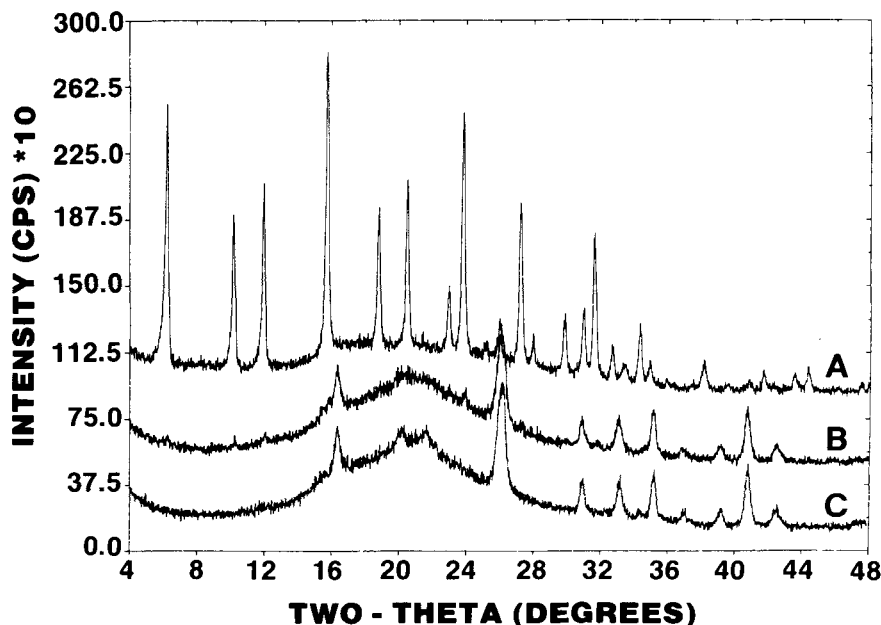


FIG. 5. X-ray diffractograms of HY crystals with 5% V calcined at (A) 540°C/10 h in air; (B) 760°C/5 h air; and (C) 760°C/5 h, steam. At 760°C the faujasite structure collapses and formation of mullite occurs.

Furthermore, a third peak near 30 ppm appears in the two HY samples under study, Fig. 4—B and 4—D; these spectra are similar to that obtained for the steam-aged gel under study. A line near 30 ppm has been attributed to the presence of highly distorted extra framework Al(IV) [30] or to pentacoordinated Al nuclei [31]. In the two RE-containing zeolites, peaks associated with Al(VI) are less intense and broader relative to the corresponding peaks seen in spectra of HY samples, indicating that steam-induced dealumination is less severe when RE ions are present. After calcination and steam-aging, extra-framework Al levels (as indicated by NMR) thus seem to decrease in the following order:

$$\text{CREY} < \text{REHY} < \text{HY(LZY-82)} \leq \text{HY (LZ-210)}$$

Extra-framework Al could complex V impurities thereby enhancing the zeolite V tolerance.

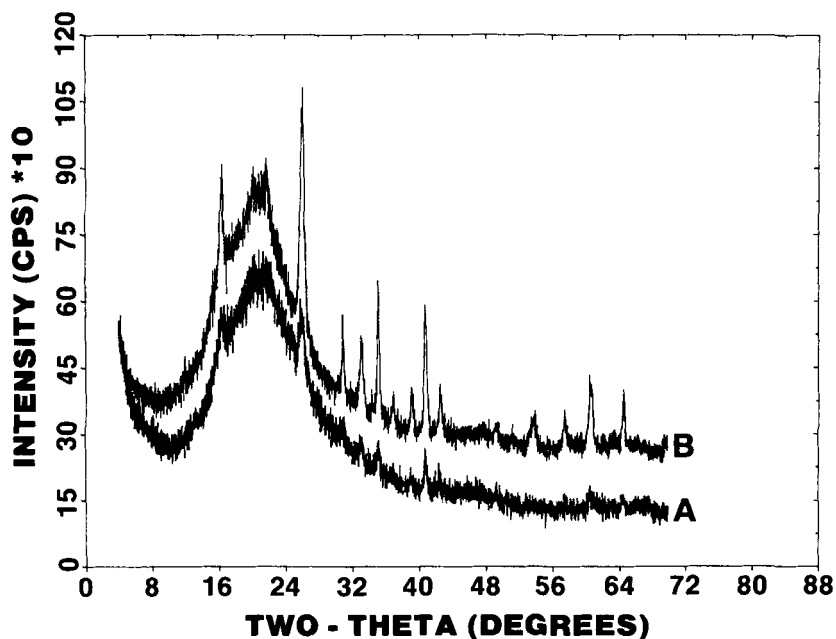


FIG. 6. X-ray diffractograms of an amorphous aluminosilicate gel after loading with 5% V and heating at: (A) 760°C in air and (B) 760°C/5 h in 100% steam.

B. Vanadium-HY Interactions

Since the Environmental Protection Agency (EPA) first proposed the lead phaseout from gasoline in late 1984, siliceous HY zeolites have replaced rare earth (RE) exchanged zeolite Y crystals in importance when manufacturing fluidized cracking catalysts. The presence or absence of lanthanide ions (together with extra-framework Al and Na⁺ ions) may affect V-zeolite interactions as well as the hydrothermal stability of V-loaded zeolites. HY crystals (with framework Si/Al = 5.3) are thermally stable when heated in air at 760°C/5 h even in the presence of 3–4% V [11]. With 5% V, the crystals are stable at 540°C (Fig. 5—trace A); however, when the calcination temperature is raised to 760°C, the faujasite structure collapses, and some mullite is formed, Fig. 5—B. Thus, even in the absence of steam, V-loaded Y zeolites collapse with an ease dependent largely on calcination temperatures and V

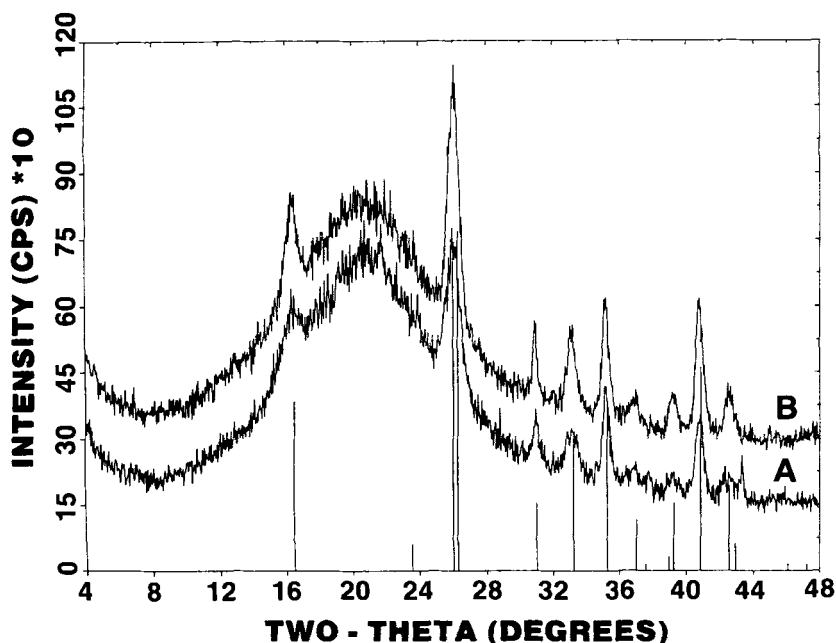


FIG. 7. X-ray diffractograms of an amorphous aluminosilicate gel (A) and HY (B) crystals calcined at 1200°C/2 h in air.

levels. In the presence of steam, the ease with which V destabilizes the faujasite lattice increases drastically. In fact, after the steam-aging (760°C/5 h) step, HY crystals with only 2% V lose their crystallinity, and formation of mullite (and tridymite) occurs [11].

Mullite formation can be observed also after calcining (760°C/5 h, air) an amorphous aluminosilicate gel (AAA-alumina) containing 5% V [11]. As observed with HY crystals, steaming (760°C/5 h) enhances mullite formation, Fig. 6. In the 1% to 5% V range, mullite formation in steam-aged gels increases with V levels. Without V, HY crystals (and the gel) form mullite only when heated at much higher temperatures. The x-ray diffractograms in Fig. 7 were obtained after calcining in air at 1200°C/2 h. Thus, it appears that V acts as a mineralizing agent decreasing by more than 400°C the temperature needed to form this mineral.

The silicon-enriched HY crystals (Linde's LZ-210) used appear to be more

TABLE 1
Unit Cell Dimensions for Several Mullites^a

Sample	Cell edge			V (a.u. ³)
	a (a.u.)	b (a.u.)	c (a.u.)	
HY at 1200°C, air	7.567	7.686	2.884	167.73
HY at 760°C, 5% V, air	7.576	7.698	2.892	168.66
HY at 760°C, 5% V, steam	7.585	7.702	2.890	168.85
Gel at 1200°C, air	7.550	7.690	2.888	167.68
Gel at 760°C, 5% V, steam	7.578	7.698	2.887	168.41
3Al ₂ O ₃ · 2SiO ₂ ^b	7.5563	7.6874	2.8842	167.54
2Al ₂ O ₃ · SiO ₂ ^b	7.5686	7.6865	2.8858	167.89

^aThe presence of an amorphous residue in these samples may affect the accuracy of the measured cell sizes, and the uncertainty of these data may be significantly higher than that associated with well-crystallized mullites free of phase impurities [33].

^bFrom Ref. 19.

resistant to V attack at hydrothermal conditions than HY (Linde's LZY-82). In fact, with 2% V, these crystals retained 40% of their initial surface area (164 m²/g) and crystallinity; 3% V is required to fully collapse the faujasite structure and to form some mullite, Fig. 8. The high extra-framework Al level present in these crystals could react with VO²⁺ (or VO₂⁺) cations and form [2 > Al—OH + VO²⁺ = (>Al—O)₂-VO + 2H⁺] complexes capable of sequestering V, thus enhancing the crystals' V resistance. In addition to mullite (and sometimes tridymite) an (x-ray amorphous) V₂O₅ phase can be seen in the laser Raman (LR) spectra for V-loaded HY crystals, Fig. 9—A. The bands at 997, 710, 530, 485, and 410 cm⁻¹ in Fig. 9—A are typical of a supported V₂O₅ phase. These bands broaden considerably for V-loaded Si-enriched HY, indicating that when supported on these crystals (after steam aging), V is probably in a highly distorted octahedral environment, Fig. 9—B.

Mullite is an aluminosilicate consisting of chains of octahedral aluminum atoms parallel to the *c* axis crosslinked by silicon or aluminum atoms in tetrahedral coordination [32]. A variety of synthetic mullites can be prepared from solid solutions (melts) of silica and alumina. The silica-rich (3Al₂O₃ · 2SiO₂) and alumina-rich (2Al₂O₃ · SiO₂) phases are called α- and β-mullite,

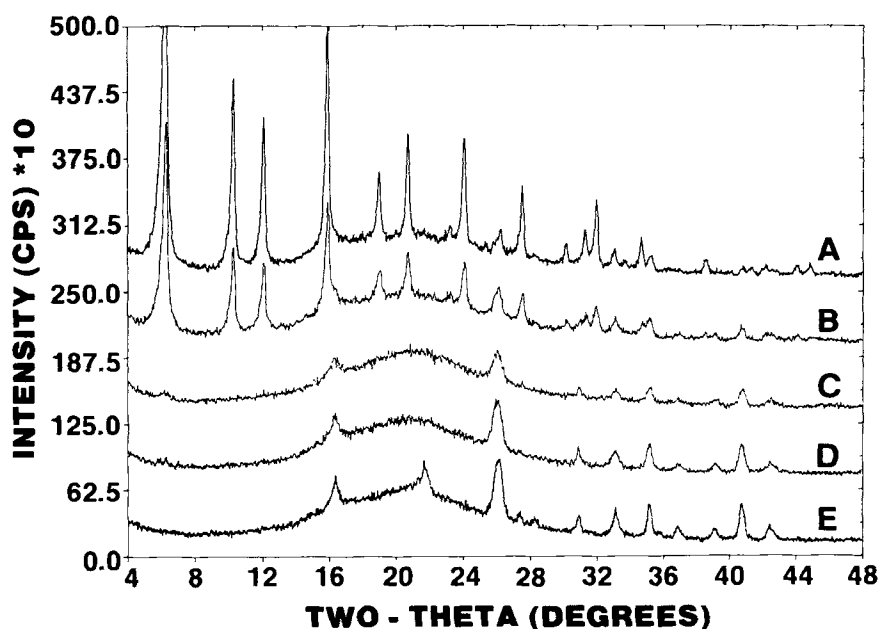


FIG. 8. X-ray diffractograms of silicon-enriched HY crystals (Linde LZ-210) steam-aged in the presence of: (A) 1.25%, (B) 2.0%, (C) 3.0%, (D) 4%, and (E) 5% vanadium.

respectively [32]. During synthesis, iron, titanium, and manganese [32] can be incorporated into the lattice [32, 33]. Iron- and titanium-bearing natural mullite are also known [34]. Substitution of aluminum in mullites can cause an expansion of the unit cell parameters and cell volume [32, 33]. However, in the case of Mn-substituted mullite, distortion of the unit cell dimensions did not change the crystals' unit cell volume [35].

The calculated orthorhombic unit cell parameters for several mullites, crystallized with and without V, are shown in Table 1. Efforts to remove the amorphous residue with HF/HCl solutions (and improve sample crystallinity) were not undertaken to avoid V leaching from the crystals. The cell dimensions shown in Table 1 indicate that (in a qualitative sense) V causes an expansion of the unit cell volume resulting mainly from an increase in the a dimension. Thus, V incorporation into these mullites probably occurs mainly by substitution of Si or Al atoms in the crosslinking tetrahedra. At the high temperatures

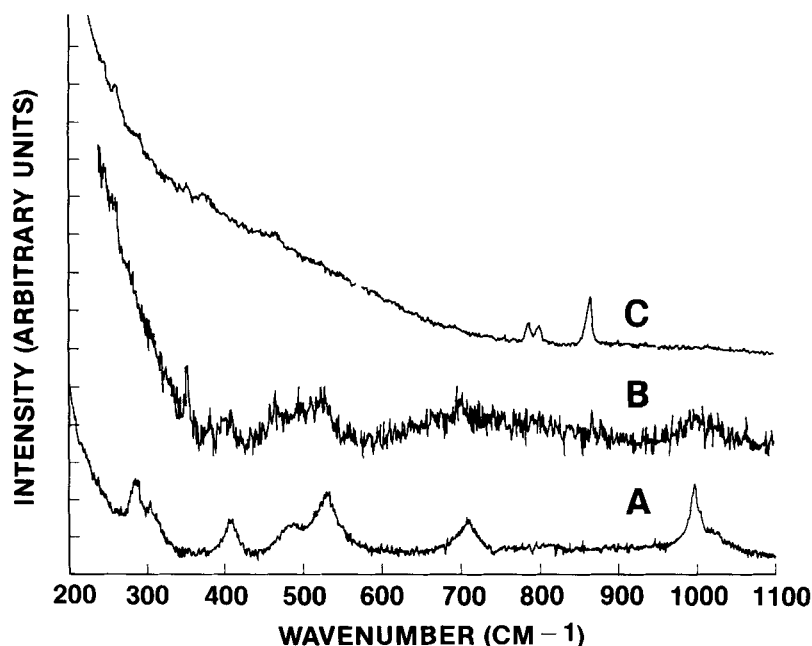


FIG. 9. Laser Raman spectra of steam-aged crystals containing 5% V: (A) HY (Linde's LZ-82), (B) silicon-enriched HY (Linde's LZ-210), and (C) REHY.

($T > 700^{\circ}\text{C}$) used in FCC regeneration, the VO^{2+} cations are probably oxidized to VO_2^+ and these oxycations could then attack the $\text{O}-\text{Al}(\text{IV})$ bond in HY. Reaction with extra-framework aluminum could then form $\text{Al}-\text{O}-\text{V}$ complexes that, when incorporated into the mullite structure cause the expansion along the a axis, and the increase in the unit cell volume shown in Table 1. Excess V forms V_2O_5 ; in samples containing more than 2% V, V_2O_5 can be easily identified by LRS, Fig. 9. In the presence of steam, hydroxovanadyl ions ($\text{VO}^{2+} + \text{H}_2\text{O} = \text{VO}(\text{OH})^+ + \text{H}^+$) and other protonated species resulting from hydrolysis of supported V_2O_5 or VO_2^+ ions, could accelerate Al removal and lattice collapse, thus increasing the ease of mullite formation indicated in Fig. 5 and elsewhere [11]. Electronic interactions between vanadyl species and framework Al in HZSM-5 have been discussed by Kucherov and Slinkin [36]. Oxygenation kinetics of VO^{2+}

cations on $\delta\text{-Al}_2\text{O}_3$ in the presence of water (at 25°C) have been reported by Wehrli and Stumm [37].

C. Interactions of Supported Vanadium with Water

During the accelerated steam-aging procedure used in the laboratory to reduce the properties of newly prepared catalysts to those of equilibrium FCC, intraparticle transport of V has been shown to occur [10]. Vanadium mobility has also been observed at pilot plant test conditions [38]. Furthermore, Kugler and Leta [39], using data from imaging secondary ion mass spectrometry (SIMS), have observed recently that an equilibrium FCC containing alumina granules accumulates V preferentially on the alumina particulates, thus supporting laboratory and pilot plant results that have indicated that V-contaminated surfaces can react with steam and from volatile compounds capable of migrating from particle to particle.

In order to obtain information that could help provide some insight on the nature of the interaction between supported V compounds and steam, several V-loaded material have been allowed to react with water and the nature of the soluble vanadate formed investigated using vanadium-51 NMR.

Amorphous V_2O_5 is thought to be a three-dimensional network of chains containing V—O linkages. At room temperature this oxide sorbs water, forming a viscous gel and, in contrast to crystalline V_2O_5 , it readily dissolves in water [40]. The nature of the species formed in V-containing solutions is still somewhat controversial [41–48], see Fig.10. In strongly alkaline solutions, VO_4^{3-} ions are present [41, 45, 46]. As the pH of the solution is decreased, HVO_4^{2-} ions are formed, which then condense to form stable $\text{V}_2\text{O}_7^{4-}$ ions ($2\text{HVO}_4^{2-} = \text{V}_2\text{O}_7^{4-} + \text{H}_2\text{O}$; $\log K = +1.68$). In the 9.5–11.9 pH range, $\text{V}_2\text{O}_7^{4-}$ is protonated to form $\text{HV}_2\text{O}_7^{3-}$ ions [41]. At lower pH, the exact nature of the vanadate species present in solution is uncertain. Brito [48] has proposed that $\text{V}_3\text{O}_3^{3-}$, $\text{V}_4\text{O}_{12}^{4-}$, $\text{HV}_6\text{O}_{16}^{3-}$, $\text{HV}_{10}\text{O}_{28}^{5-}$, and $\text{H}_2\text{V}_{10}\text{O}_{28}^{4-}$ ions are the main species present in the 2–8 pH range at 40°C.

By adding (at room temperature) small amounts of (calcined) V-contaminated materials to water (0.25 g/10 cc H_2O) after periods of 5–10 h, the hydrolysis of surface V generates liquids having different pH values and coloration. The pH of liquid containing HY (LZY-82 grade) and a dealuminated HY (LZ-210) crystals decreased from 5.8 to 3.6 and 4.4, respectively, when the V level on the zeolite increased from 0.5% to 5.0%, imparting a yellowish coloration to the liquid. The pH of liquids containing REHY crystals, by contrast, decreased only to 6.0 from 6.3 over the same V concentration

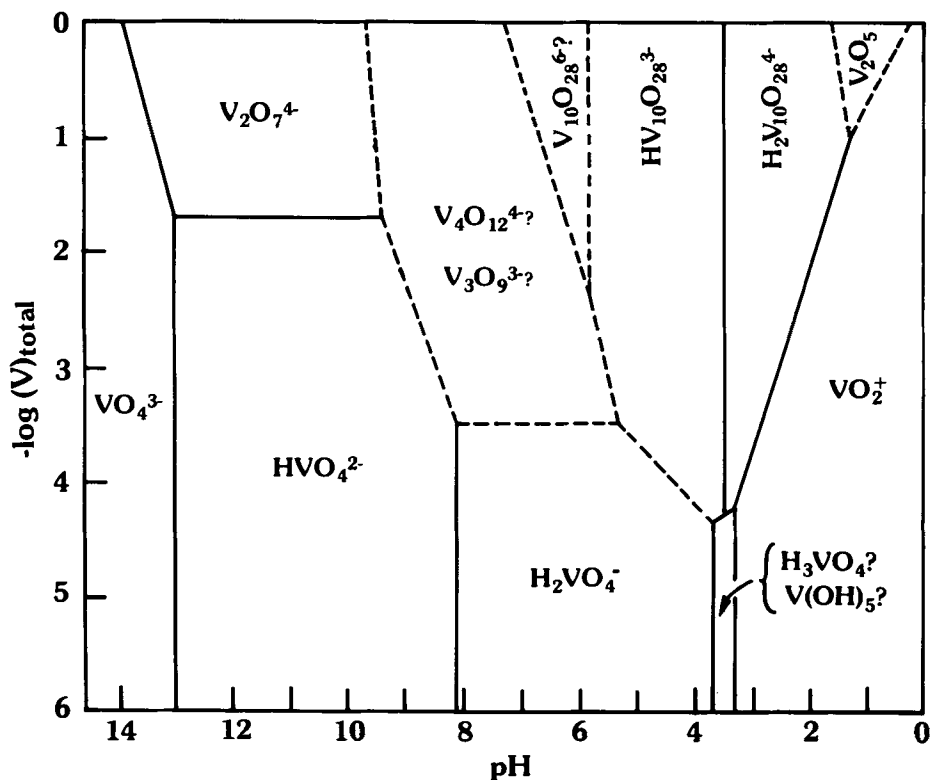


FIG. 10. Vanadate species in aqueous solution as a function of pH and total vanadium concentration (in g-atom/L) prepared by Pope and Dale [41]. Dashed lines represent uncertain demarcations.

range. Similarly prepared CREY crystals also gave solutions in the 6.0–6.7 pH range.

The ^{51}V NMR spectra for several V-containing solutions are shown in Figs. 11–14. The chemical shifts observed in these figures, together with published results [42, 43, 45], allow the identification of the major diamagnetic V^{5+} specie present. At low (20 mg V/L) V concentration, ^{51}V NMR spectra of these liquids exhibit a major resonance in the -555 to -561 ppm region (attributed to H_2VO_4^-), A traces in Figs. 11–13. Minor resonances near -570 ppm (attributed $\text{H}_2\text{V}_2\text{O}_7^{2-}$) can be seen in liquids exposed to RE-containing zeolites, Fig. 13—A. As the V concentration of the solids increases, a proportional increase in the liquid total V content (V_l) is observed and new signals in the -410 to -530 ppm region appear, indicating polymerization of VO_4 and V_2O_7 to $\text{V}_{10}\text{O}_{28}$ units. Signals in the $\sim 2:4:4$ ratio near -425 ppm, -505 ppm, and -525 ppm (and the liquid deep orange coloration) indicate the presence of a decavanadate specie (probably $\text{H}_2\text{V}_{10}\text{O}_{28}^{4-}$, Figs. 11—F, 12—D, 13—D. Additional signals near -575 ppm and near -565 ppm (attributed to $\text{HV}_2\text{O}_7^{3-}$) are also present, see Figs. 12—D and 13—D. Spectra of liquids from steam-aged samples seem to contain mainly peaks characteristics of the presence of decavanadate ions, Fig. 11—F. The good signal to noise (S/N) ratio observed in solutions obtained from steam-aged samples, reflect the high amount of V present in the $+5$ oxidation state. Since ^{51}V NMR can detect only V^{5+} ions, V compounds different from those indicated in Figs. 11–14 may also be present in these solutions. Liquids from V-loaded CREY crystals gave spectra (not shown) similar to that in Fig. 13—A with a major peak near -555 ppm and a weaker one near -567 ppm. This spectrum remained essentially unchanged even when the V loading on CREY was raised to 5%.

After calcination, V-loaded metal scavengers (such as alumina, sepiolite, attapulgite) react with water to give liquids with $\text{pH} > 7$ even when loaded with as much as 5% V. A weak band near -560 ppm (attributed to H_2VO_4^-) is seen in spectra for liquid containing 5% V on Catapal SB alumina ($\text{pH} = 7.8$), Fig. 14—A. A single major peak near -579 ppm (attributed to $\text{H}_2\text{V}_2\text{O}_7^{2-}$) is seen instead in spectra for liquids containing 5% V on sepiolite or attapulgite ($\text{pH} = 6.7\text{--}7.3$), Fig. 14—C. The spectrum in Fig. 14—C is similar to the one obtained from liquid containing $\beta\text{-Mg}_2\text{V}_2\text{O}_7$ crystals or attapulgite. The pH decreased to 4.3 from 7.8 in liquid containing steamed 5% V on Al_2O_3 and formation of decavanadate ions occurred, Fig. 14—B. In contrast, the pH increased to 8.6 from 7.3 in liquid exposed to granules of steamed sepiolite containing 5% V; their ^{51}V

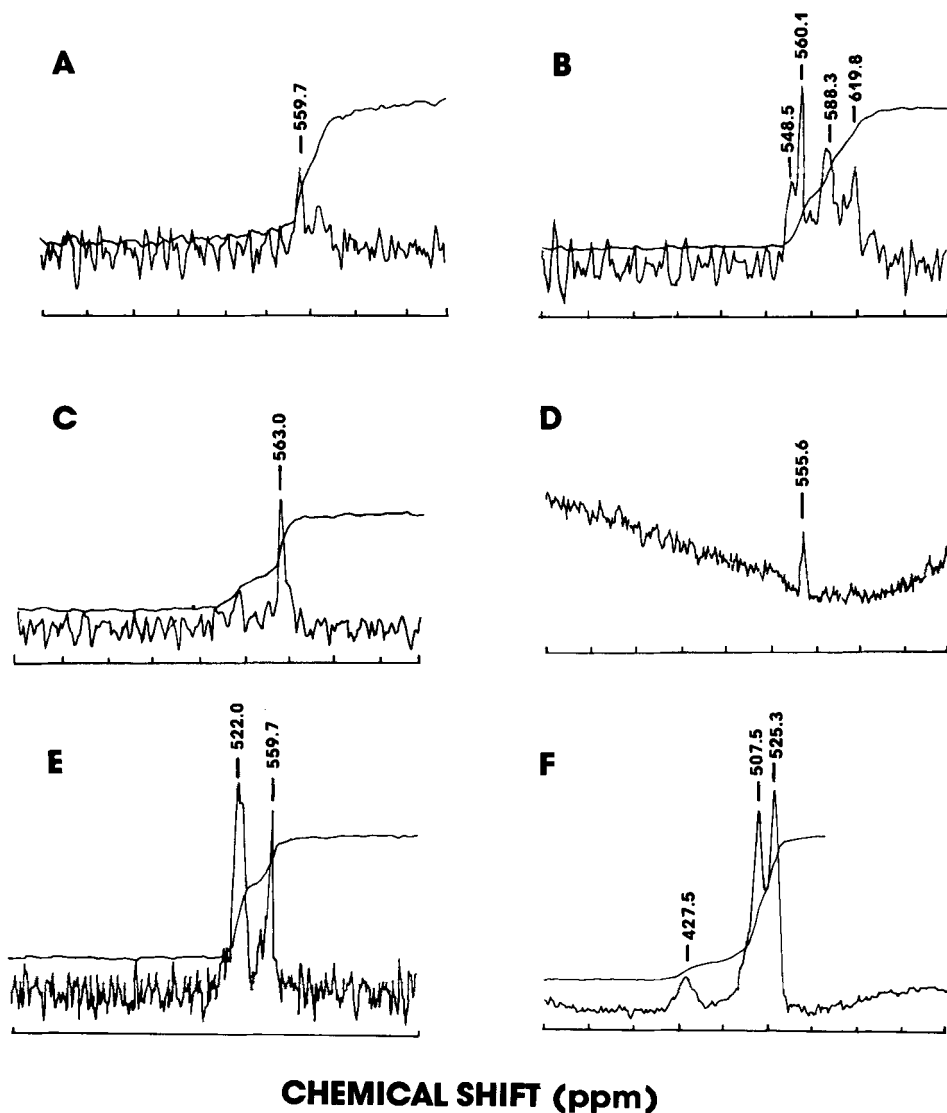


FIG. 11. Vanadium-51 NMR spectra of liquids obtained by exposing to water calcined HY crystals containing: (A) 0.4% V, (B) 0.8% V, (C) 1.0% V, (D) 2% V, (E) 5% V. F is the spectrum obtained using steamed HY with 5% V.

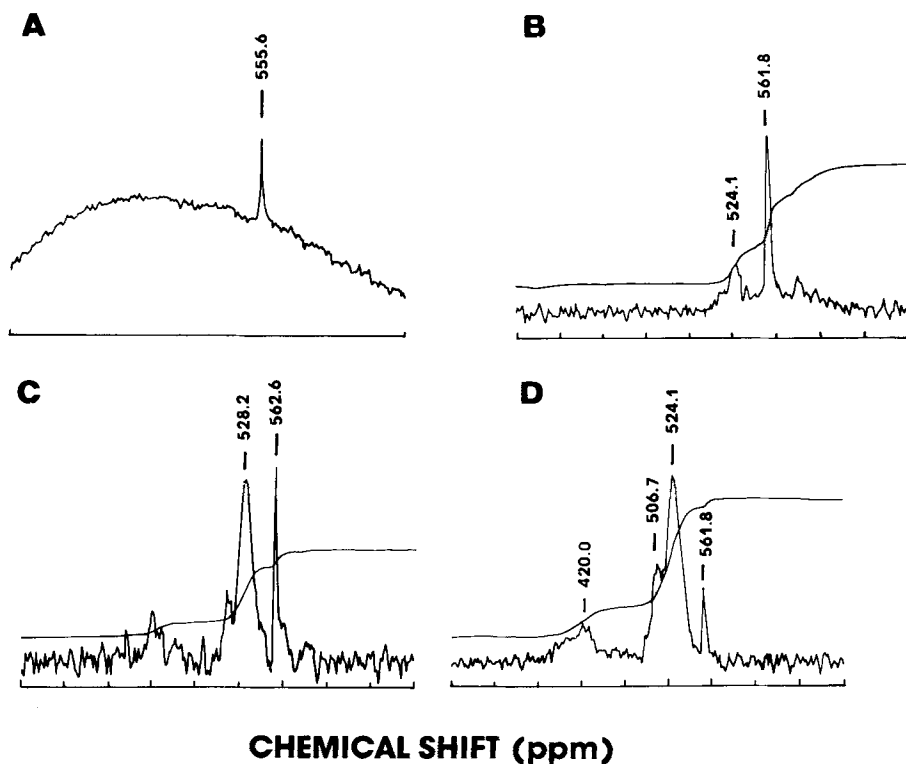


FIG. 12. Vanadium-51 NMR spectra of liquids obtained by exposing to water calcined LZ-210 crystals containing (A) 0.5% V, (B) 1.0% V, (C) 2.0% V, (D) 5.0% V.

NMR spectra exhibit a new (and sharp) resonance near -545 ppm assigned to HVO_4^{2-} species. Evidence of $\text{V}_{10}\text{O}_{28}^{8-}$ units could not be observed in liquid containing V-loaded clays, see Fig. 14. Liquids obtained by adding to water 2% V-loaded aluminosilicate gel (containing $\sim 75\%$ SiO_2 and 25% Al_2O_3) give spectra (not shown) similar to that shown in Fig. 12—A. After increasing the V level on the gel to 5%, formation of decavanadate ions occurred.

In summary, these results suggest that when exposed to water, vanadium (supported on components of FCC) undergoes complex hydrolysis-polym-

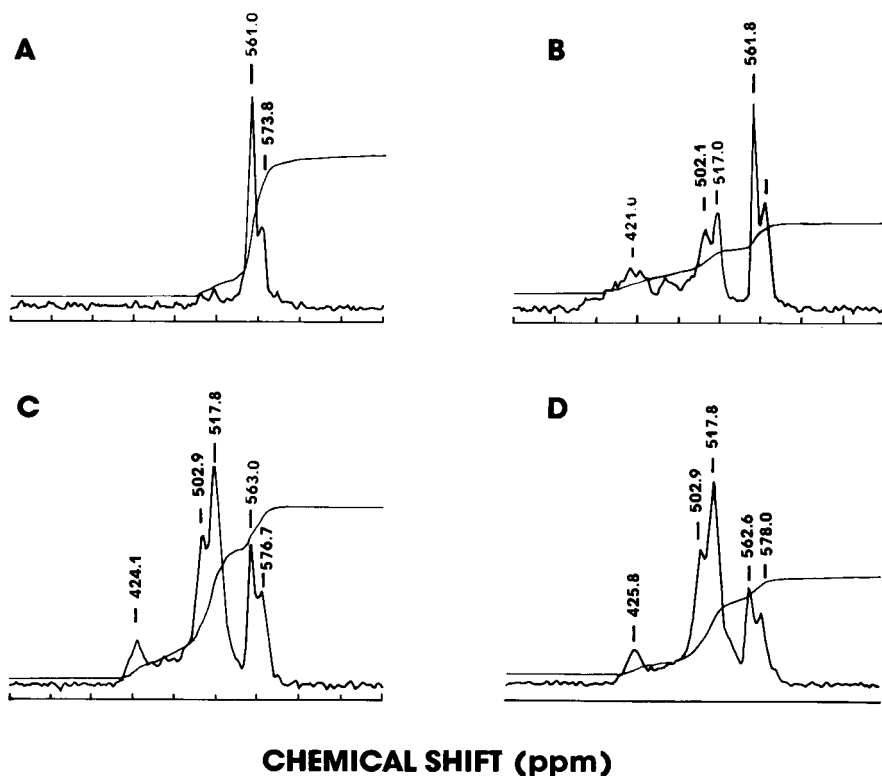


FIG. 13. Vanadium-51 NMR spectra of liquids obtained by exposing to water calcined REHY crystals containing: (A) 0.4% V, (B) 0.8% V, (C) 1.0% V, (D) 2% V.

erization reactions in which V concentration, surface composition, and liquid pH control the nature of the polyanions formed and their degree of protonation.

These data cannot predict the exact nature of the gaseous metallic hydroxides formed when a spent catalyst (V-contaminated and carbon-coated) goes from the reaction zone to the steam-stripping zone of an FCC unit. However, it is believed that the same parameters (such as surface compositions, V levels, and residence times) that influence the nature of those V compounds formed when a V-contaminated FCC is exposed to water, will also affect the nature of the volatile V compound formed when the same catalyst is exposed to steam.

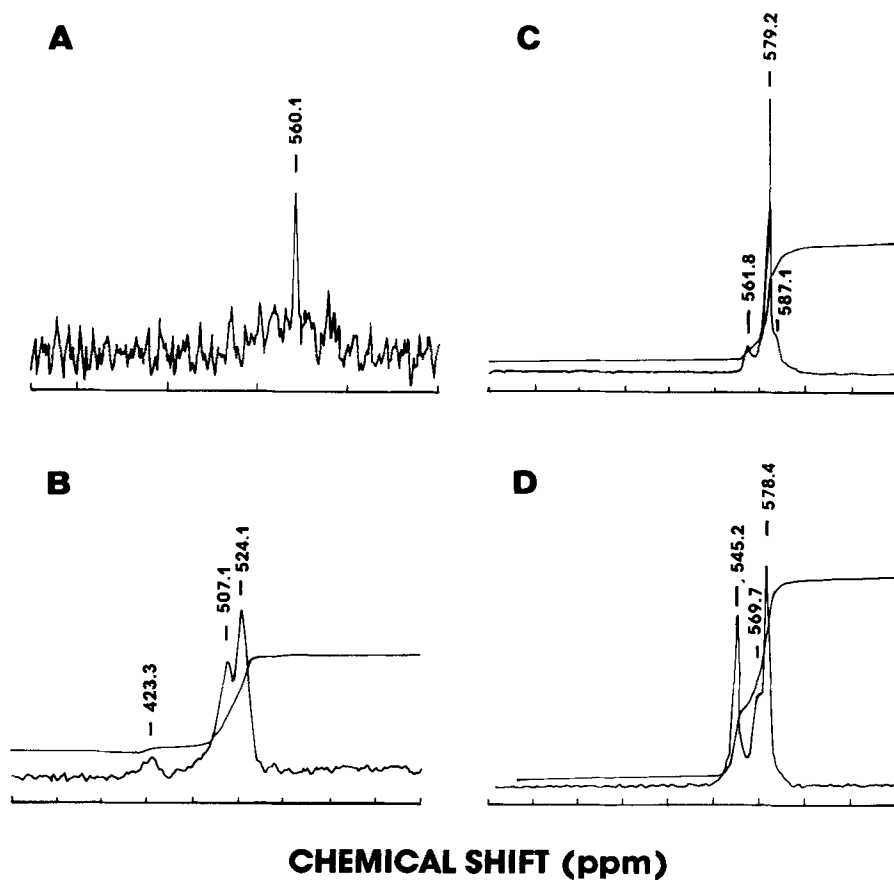


FIG. 14. Vanadium-51 NMR spectra of liquids obtained by exposing to water alumina with 5% V: (A) calcined and (B) steamed; or Sepiolite with 5% V: (C) calcined and (D) steamed.

Experimental data on the formation of volatile V compounds is scarce. The transpiration study of Glemser and Muller [49] suggest that at 527°C, the following reaction between V_2O_5 and oxygen-enriched steam occurs: $V_2O_{5(s)} + 2H_2O_{(g)} = H_4V_2O_{7(g)}$. In contrast, the transpiration study of Yannopoulos [50] has indicated that in the 327–627°C temperature range, the heterogeneous equilibrium $V_2O_{5(s,l)} + 3H_2O_{(g)} = 2H_3VO_{4(g)}$ may exist if only 1 V atom is assumed present in the gaseous molecule. Vanadic acid was believed capable of leaching Al from the zeolite framework in the form of $AlVO_4$, thus causing reduction in crystallinity and therefore cracking activity [29]. However, experimental evidence for $AlVO_4$ formation was not presented. This vanadate is not thermally stable at test conditions. In fact, the DTA profile of a reference (unsupported) $AlVO_4$ has indicated that this compound begins to decompose at about 650°C [12].

Transport experiments at microactivity test conditions have shown that in dual function FCC (DFCC) mixtures containing sepiolite, V is stabilized on the metal scavenger mainly as $\beta\text{-Mg}_2V_2O_7$ [51]. Furthermore, formation of $[V_2O_7]$ units was also observed in a steam-aged DFCC when alumina was used in place of sepiolite particles [12]. These results support the study of Glemser and Muller [49] and their conclusion that the major product of a high-temperature reaction between $V_2O_{5(s,l)}$ and steam is most likely $H_4V_2O_7$.

D. Vanadium-REY Interactions

Vanadium-loaded CREY and HY crystals have similar thermal stability (at 760°C/5 h, air) and 5% V loading is required to collapse the crystal structure. A new phase then appears that gives an x-ray diffractogram in excellent agreement with JCPDS pattern No. 12-757 for $CeVO_4$. Vanadate formation in the calcined sample can be readily identified by LRS, see Figs. 9—C and 15—A. Although the CREY crystals used contained 4% La_2O_3 , La compounds could not be observed by XRD nor by LRS [11]. In the presence of steam (at 760°C) and V, these commercially available CREY crystals are less stable than HY crystals; with 0.4% V there is a total loss of crystallinity and surface properties [11]. Formation of $CeVO_4$ can be observed near or above ~1.0% V by XRD (and LRS) in steamed samples, Fig. 15—B; vanadate formation increases with V loading [11]. As observed in a calcined samples, La-compounds were not detected. However LaY crystals containing 8.1% La_2O_3 (when steam aged in the presence of V) collapse with formation of a phase that gives a LR spectrum similar to that for $LaVO_4$ prepared by heating to 800°C a $La_2O_3\text{—}V_2O_5$ mixture, see Fig. 16.

REHY (containing 0.16% Na_2) crystals are more resistant to V attack (at

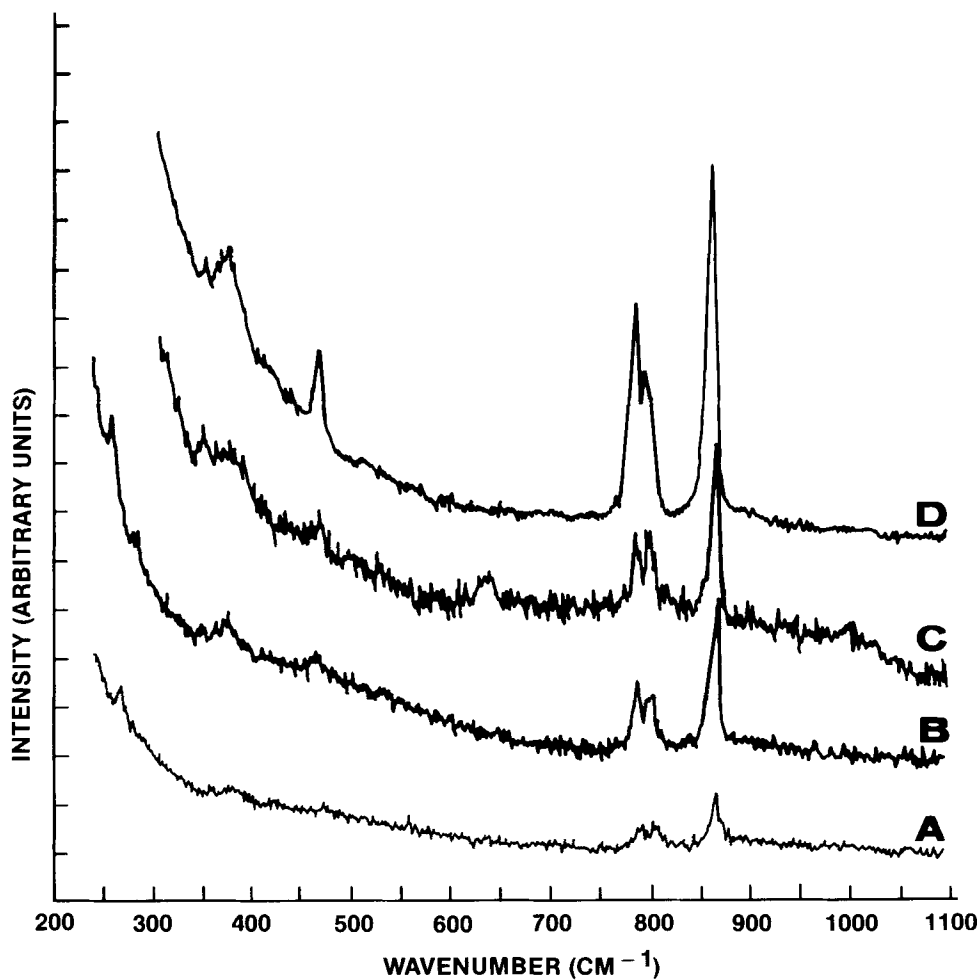


FIG. 15. Laser Raman spectra of several materials loaded with 5% V and steam-aged: (A) REHY, (B) CREY, (C) Davison's GRZ-1, a commercial FCC. The spectra for the reference unsupported CeVO_4 is shown in D.

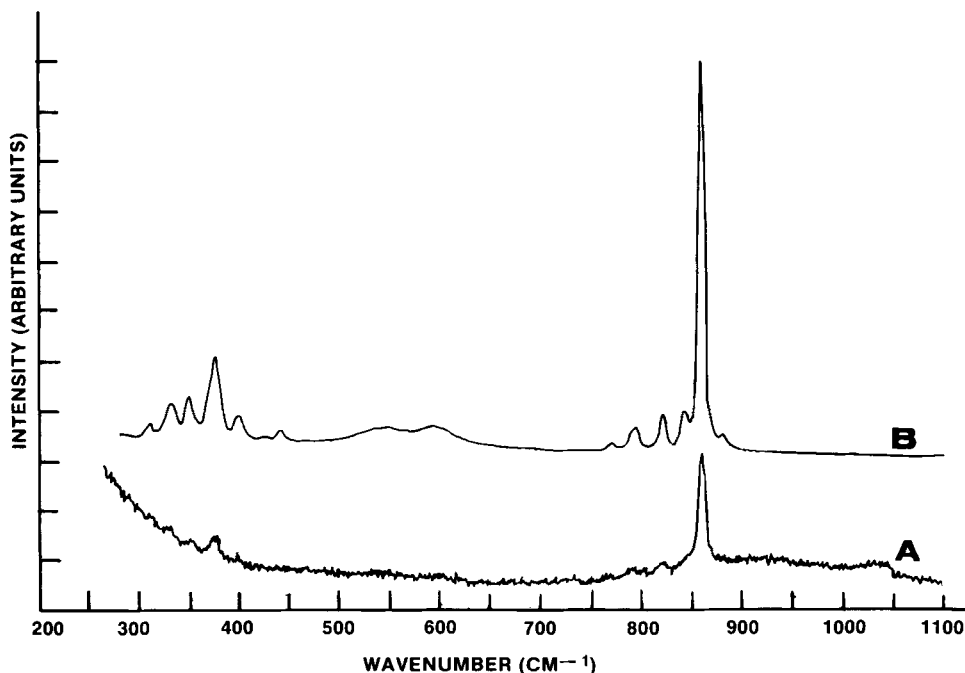


FIG. 16. Laser Raman spectra of (A) LaY with 5% V, steamed; and (B) reference LaVO_4 .

hydrothermal conditions) than CREY crystals containing 3.5% Na_2O . Vanadium could react with charge compensating Na^+ ions (to form the corresponding vanadate), thus contributing to the destabilization of the faujasite structure. With 0.5% V, REHY retains about 50% of its initial surface area ($372 \text{ m}^2/\text{g}$) and crystallinity after steam aging, Fig. 17. In the 0.5–1.0% V range, there is a drastic reduction in surface properties and at 1.25% V, evidence of the faujasite structure is lost, Fig. 17. Then, at 2% V, growth of CeVO_4 can be observed, Fig. 17.

In RE-containing Y crystals (and in REY containing FCC) it is believed that Ce^{4+} , present as an oxycerium complex is preferentially loaded near the

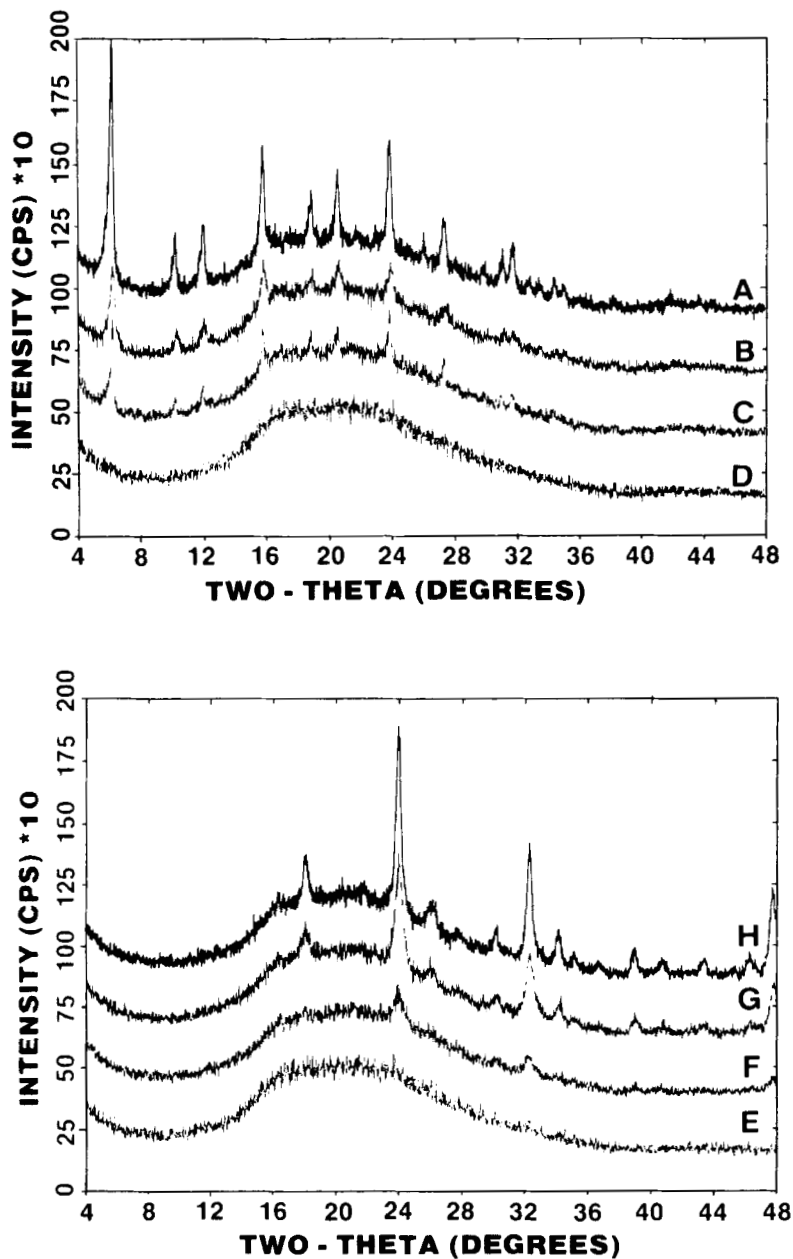


FIG. 17. X-ray diffractograms of REHY crystals steam-aged in the presence of (A) 0.5%, (B) 0.75%, (C) 1.0%, (D) 1.25%, (E) 2.0%, (F) 3.0%, (G) 4.0%, and (H) 5% vanadium.

supercages where it can more readily react with the stable $(V=O)^{2+}$ cations and form $CeVO_4$ by the following redox reaction:



Removal of lanthanides (and charge-compensating Na^+ ions) and vanadate formation, together with the enhanced framework dealumination owing to the acidity generated during steaming, are probably the main causes that can be used to rationalize the rapid and total collapse of the CREY crystals when steam-aged in the presence of V impurities.

Laser Raman spectroscopy is diagnostic of vanadate formation in CREY crystal containing more than 0.7% V. The LRS spectrum in Fig. 15—B exhibits bands near 869, 803, 786, and 370 cm^{-1} seen for unsupported $CeVO_4$, Fig. 15—D; the band near 470 cm^{-1} could be due to unreacted CeO_2 . These spectral features are retained in a V-contaminated commercial FCC (Davison GRZ-1) containing an estimated 35% CREY, Fig. 15—C. Furthermore, in the V range investigated, there is a linear dependence between the VO_4 stretching vibrational band near 870 cm^{-1} and V levels in both the CREY and the CREY-containing FCC, Fig. 18—A and 18—B. Thus, Ce–V interaction continues to occur even after the zeolite structure collapse, and $CeVO_4$ formation is not inhibited by crystallinity losses. In samples containing less than 1% V, XRD and LRS cannot identify $CeVO_4$. Nonetheless, it is believed that vanadates formation is not dependent on VO^{2+} concentration. Furthermore, the rapid collapse at 760°C (in the presence of steam) of CREY crystals with only 0.4% V is attributed mainly to vanadate formation, owing to the ease with which V can interact with the RE ions and other charge compensating cations (such as Na^+) present.

IV. THE PRESERVATION OF ZEOLITE CRYSTALLINITY AND CRACKING PROPERTIES

At the microactivity test (MAT) conditions used to test newly prepared FCC, transport experiments have shown that steam-aging induces intraparticle transport of V [10]. Addition of certain diluents to the FCC microspheres generated dual-function cracking catalyst (DFCC) mixtures that exhibit tolerance to high (1.0–1.5%) V levels. The ability of the additive to selectively sorb migrating V as heat-stable V compounds is thought responsible for the DFCC system's high V resistance reported in the patent literature and elsewhere [10, 51–53].

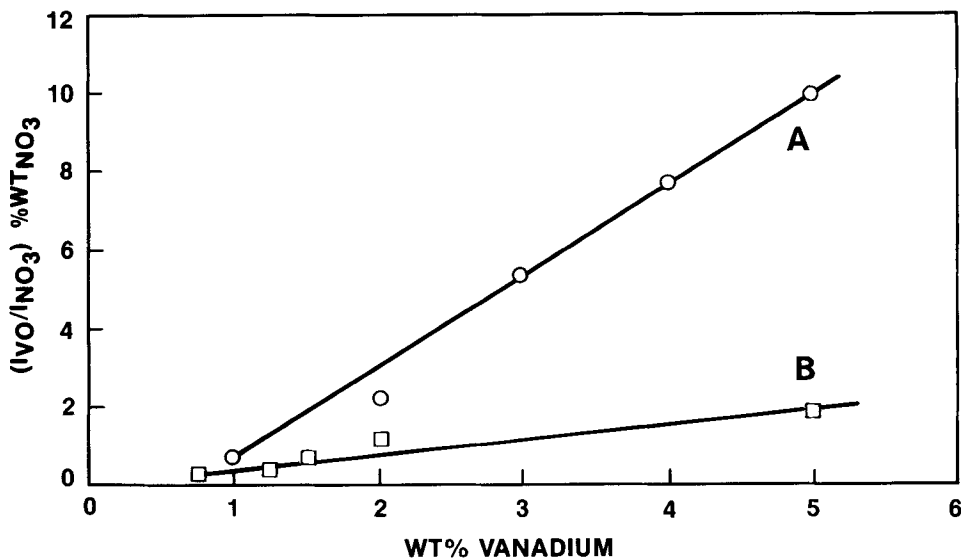


FIG. 18. The formation of CeVO_4 as a function of V loading in (A) CREY and (B) Davison's GRZ-1.

Metal scavengers reported to date have few cracking properties. Thus, when these materials are added to a commercial FCC, they act initially as diluents and decrease cracking activity, Fig. 19. However, as metals are selectively deposited on the scavenger surface, deactivation rates are significantly reduced and a cross-over point is reached at which the DFCC mixture is more active (in the presence of high metals) than the parent (undiluted) FCC, see Fig. 19. Cross-over points depend on gas oil composition, feed metal levels, zeolite concentration (and type) in the host FCC, and on the properties of the diluent. At the MAT conditions used to test FCC, DFCC mixtures containing sepiolite or aluminas, this cross-over point was observed for V loading in the 0.2–0.5% V range, Fig. 20. The activity of the zeolite-rich (35% CREY) FCC monotonically decreases with V, and for V levels in the 0.75–1.0% range it loses most of its useful cracking activity and gasoline make. However, addition of sepiolite or alumina prevents (limits) migrating V from reaching the FCC microsphere, thus preserving cracking activity. As a result, these DFCC mixtures retain a high cracking activity (70–75% conversion) and gasoline make (55–65%) even when metals loaded with as much

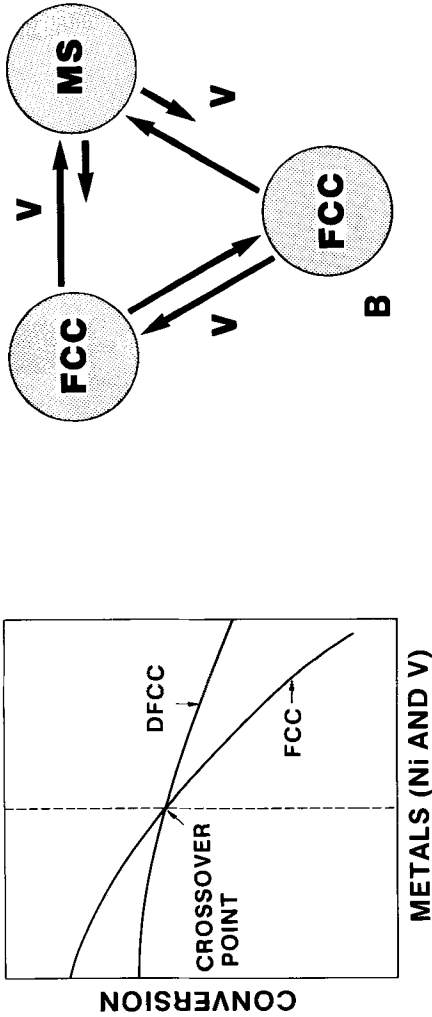


FIG. 19. The enhanced preservation of cracking activity in a DFCC mixture (A) is attributed to the metal scavenger's (MS) ability to irreversibly sorb migrating V (B).

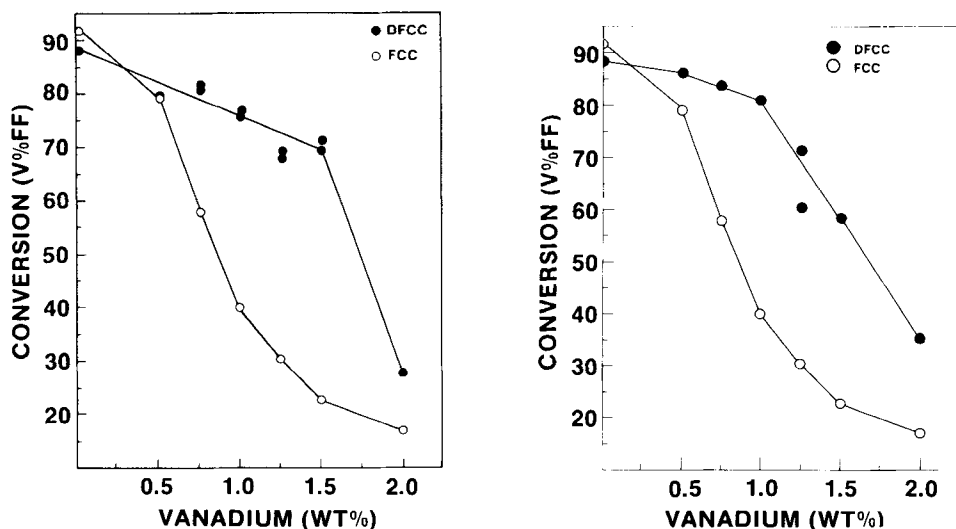


FIG. 20. An FCC retention of cracking activity in the presence of V after addition of a metal scavenger: (A) Sepiolite and (B) alumina [7, 8].

as 1.0–1.5% V; see Figs. 20 and 21. The corresponding selectivity plots are shown in Fig. 22. As the catalyst activity increases (following reduction in V levels), the heavy fractions (SO and LCGO) are cracked to produce more gasoline and light gases. Thus, liquid products distribution (and light gas make) changes in a manner controlled by catalyst activity and not by selectivity effects due to the presence of V-loaded sepiolite or alumina granules. However, the metal scavenger seems to affect the DFCC carbon (and hydrogen) make [10–12].

In cracking catalysts (with or without V), plots of coke/conversion ratio versus conversion gives curves that monotonically increase with cracking activity [51, 54, 55], see Fig. 23—A. In contrast, DFCC generate selectivity plots in which the hydrogen make and the coke/conversion ratios decrease with increasing conversion levels, that is, when V concentration levels on the FCC surface are decreased, Figs. 23—B—23—C [10–12]. During steam-aging, V preferentially migrates to the metal scavenger particles [10]. Although materials used as metal scavengers have negligible cracking activity, when loaded with V they could form dehydrogenation centers, thus converting

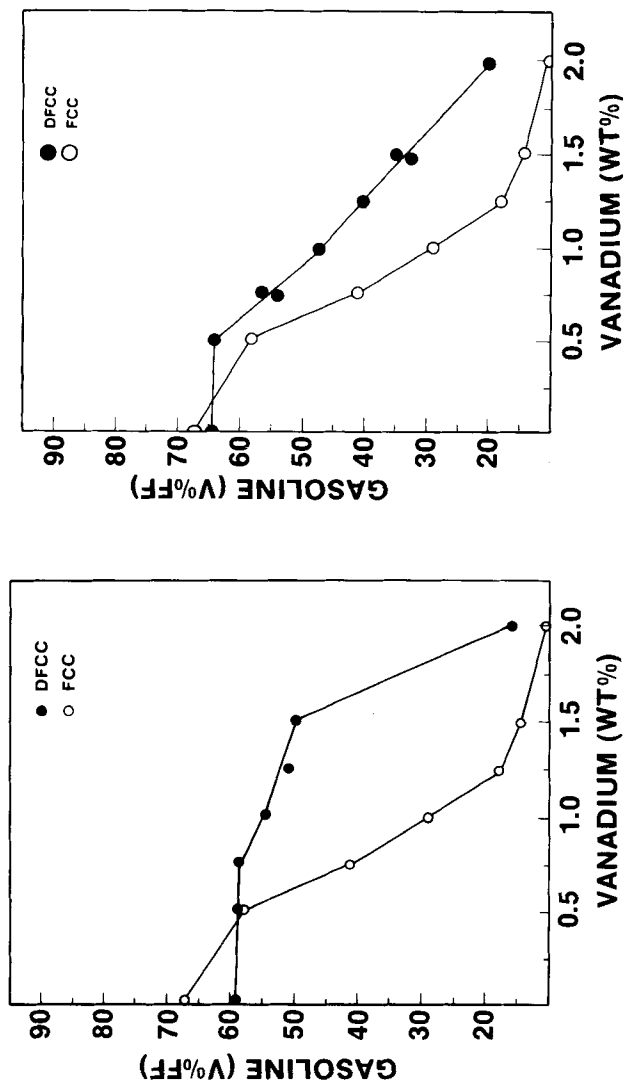


FIG. 21. An FCC retention of gasoline make in the presence of V after addition of a metal scavenger: (A) Sepiolite and (B) alumina [7, 8].

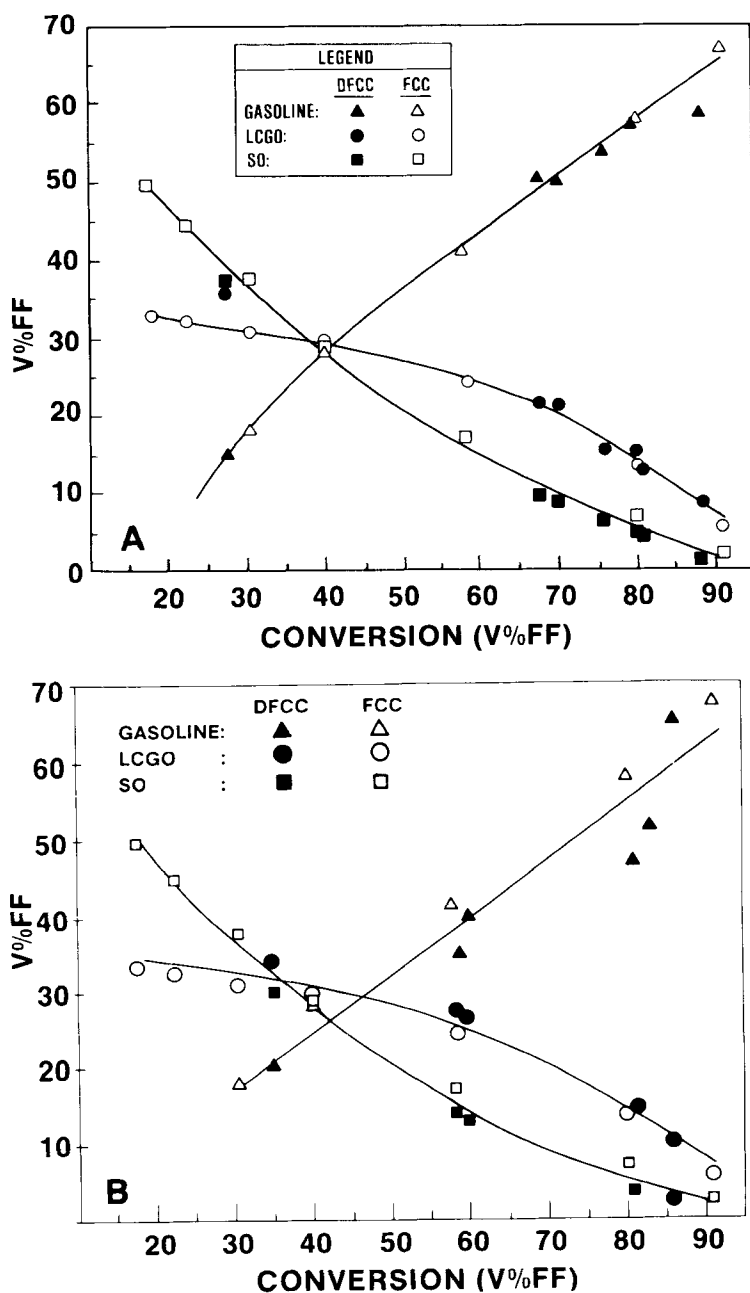


FIG. 22. Vanadium effects on liquid product selectivity in DFCC mixtures containing: (A) Sepiolite and (B) alumina [7, 8].

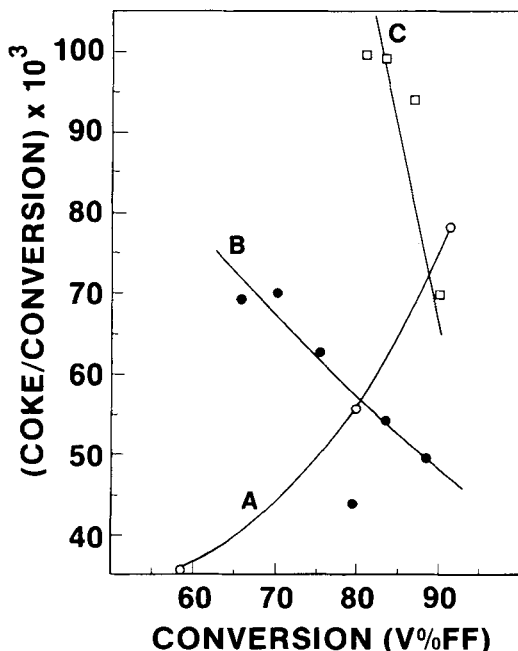


FIG. 23. Vanadium effects on carbon generation before (A) and after addition of a metal scavenger: (B) Sepiolite and (C) alumina [7, 8].

into coke and hydrogen, components that could be upgraded to gasoline-range hydrocarbons [10]. In fact, when loaded with 0–2% V, the sepiolite cracking activity remained unchanged (30% conversion) whereas carbon and H_2 generation increased threefold. In Fig. 23, the DFCC high coke make at 1.0–1.5% V (60–70% conversion) is therefore attributed to the dehydrogenation activity of the V-enriched metal scavenger particles present. The higher carbon make observed when using alumina (see Fig. 23—C) is attributed to this diluent's strong Lewis acidity, which favors the sorption of aromatics that are then retained as coke [12].

V. SUMMARY AND CONCLUSIONS

In a FCC unit, the catalyst from the regenerator, containing mainly V^{5+} compounds (such as supported V_2O_5), is admixed with fresh makeup catalyst

and then recycled to the reactor zone (the riser) where gas oil cracking occurs and more V (probably as VO^{2+}) and coke are deposited on the catalyst surface. From the riser, the spent catalyst goes to the steam-stripping zone where occluded hydrocarbons are removed from the catalyst surface. In the presence of steam, at 500–550°C, V impurities form volatile V compounds and other hydrolysis products which can react with the components (the matrix, clay, and zeolite) of an FCC, thus reducing catalyst activity and product selectivities.

In the hope of obtaining some insight into the possible nature of the reactions that may occur when V contaminated FCC are exposed to steam, the hydrolysis products of the reaction between water (at 25°C) and V contaminated (and calcined) zeolites [such as HY (Linde's LZ-82), silicon-enriched HY (Linde LZ-210), REHY, and CREY] have been examined by ^{51}V NMR. Results have indicated that formation of anions such as H_2VO_4^- and HVO_4^{2-} ; condensation reaction products such as $\text{H}_2\text{V}_2\text{O}_7^{2-}$, $\text{HV}_2\text{O}_7^{3-}$, and $\text{H}_2\text{V}_2\text{O}_7^{2-}$; and polyanions such as $\text{H}_2\text{V}_{10}\text{O}_{28}^{4-}$ depend on contact times, catalyst composition, V concentration, and the pH of the liquid examined.

During the time (30 to 60 sec) that spent catalysts spend in the steam-stripping zone, different reaction mechanisms and compounds are expected from the reaction between supported vanadium species and steam. However, it is believed that the same parameters (such as contact time, the FCC surface composition, and V loadings) that determine the nature of the vanadate formed in the liquid phase when deposited V impurities react with water, could also determine the nature of the volatile compounds (responsible for intraparticle V transport) formed during steam regeneration of spent FCC.

LRS characterization of DFCC systems, tested at MAT conditions, has indicated that in steam-aged catalysts containing more than 1% V, $[\text{V}_2\text{O}_7]$ units are formed, lending support to the experimental work of Glemser and Muller [49] in which the reaction $\text{V}_2\text{O}_{5(s,l)} + 2\text{H}_2\text{O(g)} = \text{H}_4\text{V}_2\text{O}_7\text{(g)}$ was reported. Thus it is believed that $\text{H}_4\text{V}_2\text{O}_7$ is one of the volatile V compounds that can be formed in the steam-stripping zone of an FCC unit.

^{29}Si and ^{27}Al MASNMR spectroscopy has shown that on calcination, FCC cracking components such as HY, Si-enriched HY, REHY, and CREY undergo dealumination, and that steam-aging increases the presence of extra-framework Al in these zeolites; dealumination is more severe in HY-type materials. The interactions between V impurities and the FCC cracking components (the zeolite) during the regeneration cycle of an FCCU are affected by the presence or absence of RE ions in the zeolite.

Faujasite crystals (with or without RE ions) are thermally stable at 760°C/5 h in air for V loading up to ~4%. When the V on catalyst levels are raised

to 5%, the crystal structure collapses, indicating that, at this V concentration and temperature, the presence of steam is not required to induce zeolite destruction. However, when the same thermal treatment is repeated in the presence of steam (at 1 atm), the ease with which V destroys the zeolite crystallinity greatly increases; CREY crystals (in the presence of 3.5% Na_2O) collapse with only 0.4% V. The zeolite V resistance seems to decrease when RE ions are present and to increase with increasing extra-framework Al (generated during steam-aging) in the following order:

HY (Linde's LZ-210) > HY (Linde's LZY-82) > REHY > CREY

Extra-framework Al in HY-type zeolites could directly react with VO^{2+} cations and form stable complexes, thus increasing the crystal tolerance to V.

HY-type crystals as well as x-ray amorphous aluminosilicate gels (AAA alumina), form mullite when heated at 1200°C in air. In the presence of V, these materials form mullite at lower temperatures (760°C), indicating that V acts as a mineralizing agent. In addition to mullite and tridymite, an x-ray amorphous V_2O_5 phase is also formed. Calculations of the orthorhombic unit cell parameter have indicated that an expansion in the a dimension occurs when mullite is formed in the presence of V, suggesting that the observed expansion in the unit cell volume is probably due to V incorporation into the crystal lattice. At the high temperatures used for FCC regeneration, oxyanions of vanadium (VO_2^+ or VO^{2+}) could attack the Al—O—Si bond in HY and cause lattice collapse. The amorphous product recrystallizes then into mullite and the excess silica forms tridymite; excess V will form a supported V_2O_5 phase. In the presence of steam, hydroxy vanadyl ions and other protonated species resulting from the interaction of V compounds (such as V_2O_5) with steam, could accelerate Al removal from the faujasite structure, thus increasing the ease of mullite formation.

In REHY and CREY crystals it is believed that Ce^{4+} ions present as an oxycerium complex undergo a redox reaction with oxyvanadyl cations (VO^{2+}) and form a stable orthovanadate. Removal of other charge-compensating cations (such as Na^+ ions) in the form of vanadates further destabilizes the crystal lattice, thus promoting zeolite destruction. In fact, whereas with 0.4% V CREY (containing 3.5% Na_2O) became essentially x-ray amorphous after steaming, REHY (containing 0.16% Na_2) required three times as much V for total lattice destruction. Vanadate formation, together with framework dealumination resulting from the acidity generated by the hydrolysis of V

compounds during steaming, is believed responsible for the greater ease with which V impurities can destabilize RE-containing faujasite crystals. The effects of Na impurities (and of Si/Al ratios) on the V tolerance of a zeolite with the faujasite structure have been described in a recent paper [56].

The deleterious effects of V deposits on zeolite-containing FCC can be greatly reduced by addition of certain materials (metal scavengers) capable of minimizing V-FCC interactions by selectively (and irreversibly) sorbing migrating V compounds such as $H_4V_2O_7$. Dual-function cracking catalysts (DFCC) mixtures have been prepared that can retain most of their useful cracking activity (at MAT condition) even in the presence of 1.0–1.5% V. Thus, DFCC systems appear to have the necessary metal tolerance necessary to crack residual oil as well as V-contaminated crudes and may offer cost (as well as coke selectivity) advantages over conventional FCC.

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REFERENCES

- [1] M. L. Occelli, *Fluid Catalytic Cracking: Role in Modern Refining*, ACS Symp. Ser. Vol. 375 (M. L. Occelli, ed.), ACS, Washington, DC, 1989, p. 1
- [2] G. H. Dale and D. L. McKay, *Hydroprocessing* **56**(9), 97–102 (September 1977).
- [3] R. G. Meisenheimer, *J. Catal.*, **1**, 356 (1962).
- [4] R. N. Cimbalo, R. L. Foster, and S. J. Wachtel, *Oil Gas J.*, 112 (1972).
- [5] J. S. Magee, R. E. Ritter, and L. Rheume, *Hydrocarbon Process*, 123 (1979).
- [6] R. E. Ritter, L. Rheume, W. A. Welsh, and J. S. Magee, *Oil Gas J.*, 103, (1981).
- [7] M. L. Occelli, D. Psaras, and S. L. Suib, *J. Catal.*, **96**, 336 (1986).
- [8] M. L. Occelli, D. Psaras, and S. L. Suib, in *Proc. Int. Symp. Zeo. Catal.*, Siofok, Hungary, 1985, p. 423.
- [9] M. L. Occelli and J. M. Stencel, in *Proc. 9th Int. Congress Catalysts*, (M. J. Phillips and M. Ternan, eds.), 1988, Vol. 1, p. 214.

- [10] M. L. Occelli, in *Fluid Catalytic Cracking: Role in Modern Refining*, ACS Symp. Ser. Vol. 375 (M. L. Occelli, ed.), ACS, Washington, DC, 1989, p. 162.
- [11] M. L. Occelli and J. M. Stencel, in *Zeolites as Catalysts, Sorbents and Detergent Builders*, (H. G. Karge and J. Weitkamp, eds.), Elsevier, 1989, p. 127.
- [12] M. L. Occelli and J. M. Stencel, in *Zeolites: Facts, Figures, Future*, Proc. 8th IZA Meeting, Amsterdam, The Netherlands (P. A. Jacobs and R. A. Van Santen, eds.), 1989, p. 1311.
- [13] L. F. Flanders and C. E. Rudy, U.S. Patent 3,409,541 (1968).
- [14] J. DeJong, U.S. Patent 4,519,897 (1985).
- [15] B. R. Mitchell and R. F. Vogel, U.S. Patent 4,520,120 (1985).
- [16] D. J. Rawlence, K. Gosling, L. Steel, and A. P. Chapple, in *Sci. Bases Prep. Het. Catalysts*, 5th Int. Symp. Preprints, 1990, p. 279.
- [17] M. L. Occelli, G.B. Patent 2,145,345A (1985).
- [18] W. P. Hettinger, in *Fluid Catalytic Cracking: Role in Modern Refining*, ACS Symp. Ser., Vol. 375 (M. L. Occelli, ed.), ACS, Washington, DC 1989, p. 308.
- [19] B. R. Mitchell, *Ind. Chem. Prod. Res. Dev.*, **19**, 209 (1980).
- [20] A. W. Chester, *Ind. Eng. Chem. Res.*, **26**(5), 863 (1987).
- [21] M. W. Anderson, M. L. Occelli, and S. L. Suib, *J. Catal.*, **118**, 31 (1989).
- [22] M. W. Anderson, M. L. Occelli, and S. L. Suib, *J. Catal.*, **122**, 374 (1990).
- [23] G. L. Woolery, A. A. Chin, G. W. Kirker, and A. M. Huss, in *Fluid Catalytic Cracking: Role in Modern Refining*, ACS Symp. Ser. Vol. 375 (M. L. Occelli, ed.), ACS, Washington, D.C, 1989, p. 215.
- [24] S. Jaras, *Appl. Catal.*, **2**, 207 (1982).
- [25] L. Upson, S. Jaras, and I. Dalin, *Oil Gas J.*, 135 (1982).
- [26] R. Pompe, S. Jaras, and N. Vannenberg, *Appl. Catal.*, **13**, 171 (1984).
- [27] F. Mauge, J. C. Courcelle, Ph. Engelhard, P. Gallezot, and J. Grosmangin, *Zeolites*, **6**, 261 (1986).
- [28] F. Mauge, J. C. Courcelle, Ph. Engelhard, P. Gallezot, and J. Grosmangin, in *New Developments in Zeolite Science and Technology* (Y. Murakami, A. Iijima, and J. W. Ward, eds.), Kodansha-Elsevier, Tokyo, 1986, p. 803.
- [29] R. F. Wormsbecher, A. W. Peters, and J. M. Maselli, *J. Catal.*, **100**, 130 (1986).
- [30] J. P. Gilson, G. C. Edwards, A. K. Peters, K. Rajagopalan, R. F. Wormsbecher, T. G. Roberie, M. P. Shatlock, *J. Chem. Soc. Chem. Commun.*, 91 (1987).
- [31] A. Samoson, E. Lippmaa, G. Engelhardt, D. Lohse, and H. G. Jerschwitz, *Chem. Phys. Lett.* **134**, 589 (1987).
- [32] S. O. Agrell and J. V. Smith, *J. Am. Ceram. Soc.*, 69 (1960).
- [33] P. Das, and R. Choudhury, *Am. Ceram. Soc. Bull.*, 188 (1988).

- [34] W. A. Deer, R. A. Howie, and J. Zussman, in *An Introduction to the Rock Forming Minerals*, Langman, 1973, p. 37.
- [35] H. Schneider and R. Vasudevan, *N. Jb. Mineral Mh.*, **4**, 165 (1989).
- [36] A. V. Kucherov and A. A. Slinkin, *Zeolites*, **7**, 583 (1987).
- [37] B. Wehrli and W. Stumm, *Langmuir*, **4**, 753 (1988).
- [38] M. L. Occelli, unpublished results (1982).
- [39] E. L. Kugler and D. D. Leta, *J. Catal.*, **109**, 387 (1988).
- [40] C. Sanchez, J. Livage, and G. Lucazeau, *J. Raman Spect.*, **12**(1), 68 (1980).
- [41] M. T. Pope and B. W. Dale, *Rev. (London)*, **22**, 527 (1968).
- [42] S. E. O'Donnell, M. T. Pope, *J. Chem. Soc., Dalton Trans.*, **21** (1976).
- [43] E. Heath, and O. W. Howarth, *J. Chem. Soc., Dalton Trans.*, **23** (1981).
- [44] L. R. Le Coustumer, B. Taouk, M. Le Meur, E. Payen, M. Guelton, and J. Grimblot, *J. Phys. Chem.*, **92**, 1230 (1988).
- [45] B. Taouk, M. Guelton, J. Grimblot, and J. P. Bonnelle, *J. Phys. Chem.*, **92**(23), 6700 (1988).
- [46] F. G. Vilchez and W. P. Griffith, *J. Chem. Soc., Dalton Trans.*, 1416 (1972).
- [47] W. P. Griffith and P. J. B. Lesniak, *J. Chem. Soc. A*, 1066 (1969).
- [48] F. Brito, *Acta Chem. Scand*, **21**, 1968 (1967).
- [49] O. Glemser and A. Muller, *Z. Anorg. Allg. Chem.*, **325**, 220 (1963).
- [50] L. N. Yannopoulos, *J. Phys. Chem.*, **72**(9), 3293 (1968).
- [51] M. L. Occelli and J. M. Stencel, in *Fluid Catalytic Cracking: Role in Modern Refining*, ACS Symp. Ser. Vol. 375 (M. L. Occelli, ed.), ACS, Washington, DC, 1989, p. 195.
- [52] M. L. Occelli and J. V. Kennedy, U.S. Patent 4,465,588 (1984).
- [53] M. L. Occelli and H. E. Swift, U.S. Patent 4,466,884 (1984).
- [54] R. N. Cimbalo, R. L. Foster, and S. J. Wachtel, *Symposium on New Developments in FCC*, 37th Meeting of the API Division of Refining, Preprint No. 87-72, New York, (1972).
- [55] M. L. Occelli, D. C. Kowalczyk, and C. L. Kibby, *Appl. Catal.*, **16**, 227 (1985).
- [56] M. L. Occelli, to be presented at the AKZO Catalyst Symposium, Scheveningen, The Netherlands, June 1991.